Sample and Data Relationship Format for Proteomics (SDRF-Proteomics)

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Table of Contents

1. Status of this document **1**

2. Abstract **2**

3. Motivation **3**

4. Specification structure **4**

5. The SDRF-Proteomics Format **5**

5.1. Versioning . . . 6

5.2. Format rules . . . 7

5.3. Reserved words . . . 7

5.4. SDRF file-level metadata . . . 8

5.5. Table Column headers . . . 9

5.6. Table Cell values . . . 10

6. Validating SDRF Files **11**

7. SDRF-Proteomics: Samples metadata **12**

7.1. BioSamples database integration . . . 12

7.2. Encoding sample technical and biological replicates . . . 13

7.3. Pooled samples . . . 13

7.4. Sample Metadata Guidelines . . . 14

8. SDRF-Proteomics: data files metadata **15**

8.1. CV Term Format for Data File Metadata . . . 15

8.2. Vendor formats that ship multiple files (e.g. AB Sciex .wiff + .wiff.scan) . . . 16

8.3. Sample Preparation and Fragmentation (MS-based only) . . . 17

8.4. Proteomics data acquisition method . . . 17

8.5. MS-Proteomics Template . . . 18

9. Additional SDRF Rules **19**

9.1. Column Cardinality . . . 19

9.2. Row Uniqueness Requirements . . . 19

10. Templates **20**

10.1. Template Architecture . . . 20

10.2. Template Combination Rules . . . 20

10.3. Specifying Templates in SDRF Files . . . 21

10.4. Available Templates . . . 22

10.5. Extending Templates . . . 22

10.6. Contributing New Templates . . . 22

11. Factor Values (Study Variables) **23**

11.1. Column Format . . . 23

11.2. When to Use Factor Values . . . 23

11.3. Rules . . . 23

11.4. Example . . . 23

12. Ontologies and Controlled Vocabularies **24**

13. Examples of Annotated Datasets **26**

14. Template Definitions **27**

14.1. base . . . 27

14.2. sample-metadata . . . 28
14.3. ms-proteomics . . . 30
14.4. affinity-proteomics . . . 34
14.5. human . . . 35
14.6. vertebrates . . . 35
14.7. invertebrates . . . 36
14.8. plants . . . 36
14.9. clinical-metadata . . . 37
14.10. oncology-metadata . . . 39
14.11. dia-acquisition . . . 40
14.12. single-cell . . . 41
14.13. immunopeptidomics . . . 43
14.14. crosslinking . . . 44
14.15. cell-lines . . . 46
14.16. metaproteomics . . . 47
14.17. human-gut . . . 49
14.18. soil . . . 52
14.19. water . . . 58
14.20. gc-ms-metabolomics . . . 64
14.21. lc-ms-metabolomics . . . 65
14.22. ms-metabolomics . . . 67
15. Intellectual Property Statement **71**
16. Copyright Notice **72**
17. How to cite **73**
References **74**

Chapter 1. Status of this document

This document provides information to the proteomics community about a proposed standard for sample metadata annotations in public repositories called Sample and Data Relationship Format (SDRF)-Proteomics. Distribution is unlimited.

Version v1.1.0 - 2026-01

Chapter 2. Abstract

The Human Proteome Organisation (HUPO) Proteomics Standards Initiative (PSI) defines community standards for data representation in proteomics to facilitate data comparison, exchange, and verification. This document presents a specification for the Sample and Data Relationship Format (SDRF-Proteomics).

Further detailed information, including any updates to this document, implementations, and examples is available at SDRF GitHub Repository. The official PSI web page for the document is: HUPO-PSI SDRF.

Chapter 3. Motivation

Public proteomics data is valuable, but sample metadata is often missing or stored inconsistently across repositories (e.g., CPTAC uses Excel files, ProteomicsDB captures minimal properties) [1]. This heterogeneity prevents reproducibility and cross-dataset integration.

SDRF-Proteomics addresses this by providing a standard tab-delimited format to capture (**Figure 1**):

- Sample metadata and characteristics
- Data file acquisition parameters
- Sample-to-file relationships (experimental design)

[sample metadata] | *https://github.com/bigbio/proteomics-metadata-standard/raw/master/sdrf-proteomics/images/sample-metadata.png*

Figure 1: SDRF-Proteomics captures sample information and its relationship to data files.

The format is fully compatible with MAGE-TAB SDRF, enabling integration with transcriptomics metadata standards.

Chapter 4. Specification structure

SDRF-Proteomics uses a two-tier system: this **core specification** defines the format rules, and **templates** provide metadata checklists for specific experiment types (**Figure 2**). Templates are organized in the templates/ directory, each with documentation and example files.

[Logo] | *images/sdrf-guidelines-structure.png*

Figure 2: SDRF-Proteomics specification structure. The main specification defines the core rules and is extended by sample templates (human, vertebrates, etc.) and experiment-type templates (crosslinking, immunopeptidomics, etc.).

The official repository is GitHub, where you can find annotated example projects and the official validator sdrf-pipelines.

IMPORTANT	Throughout this specification, the keywords "MUST", "REQUIRED", "SHOULD", "RECOMMENDED", and "OPTIONAL" are interpreted as described in RFC 2119.

Chapter 5. The SDRF-Proteomics Format

SDRF-Proteomics is a **tab-delimited file** where:

- Each **row** = one sample linked to one data file
- Each **column** = a property (sample characteristic, data file attribute, or factor value)
- Each **cell** = the property value for that sample/file or a factor value.

Here's a minimal example:

source name	characteristics[organism]	characteristics[organism part]	characteristics[disease]	characteristics[biological replicate]	assay name	technology type	comment[proteomics data acquisition method]	comment[label]	comment[instrument]	comment[cleavage agent details]	comment[fraction identifier]	comment[technical replicate]	comment[data file]	factor value[disease]
sample_1	homo sapiens	liver	normal	1	run_1	proteomic profiling by mass spectrometry	data-dependent acquisition	label free sample	Q Exactive HF	NT=Trypsin;AC=MS:1001251	1	1	sample_1.raw	normal
sample_2	homo sapiens	liver	hepatocellular carcinoma	1	run_2	proteomic profiling by mass spectrometry	data-dependent acquisition	label free sample	Q Exactive HF	NT=Trypsin;AC=MS:1001251	1	1	sample_2.raw	hepatocellular carcinoma

source name	characteristics[organism]	characteristics[organism part]	characteristics[disease]	characteristics[biological replicate]	assay name	technology type	comment[proteomics data acquisition method]	comment[label]	comment[instrument]	comment[cleavage agent details]	comment[fraction identifier]	comment[technical replicate]	comment[data file]	factor value[disease]
sample_3	homo sapiens	not available	not available	1	run_3	proteomic profiling by mass spectrometry	data-dependent acquisition	label free sample	Q Exactive HF	NT=Trypsin;AC=MS:1001251	1	1	sample_3.raw	not available

The file is organized into three column sections:

1. **Sample metadata** (`characteristics[…]`) - organism, disease, tissue, etc.
2. **Data file metadata** (`comment[…]`) - instrument, label, fraction, data file
3. **Factor values** (`factor value[…]`) - variables under study for statistical analysis

NOTE

- This example shows **mass spectrometry** proteomics - see MS-Proteomics template for full requirements.
- For **affinity proteomics** (Olink, SomaScan), see Affinity-Proteomics template.
- Unknown values use reserved words: `not available`, `not applicable`, or `pooled`.
- For a step-by-step tutorial, see the Quick Start Guide.

5.1. Versioning

The SDRF-Proteomics specification uses Semantic Versioning (MAJOR.MINOR.PATCH). Version numbers are prefixed with "v" (e.g., v1.1.0). Changes are proposed via GitHub pull requests to the dev branch.

For the complete versioning strategy — including template versioning, ontology updates, the deprecation policy, transition timelines, and migration tooling — see Versioning and Deprecation Policy.

5.2. Format rules

- **Case sensitivity**: Text values are case-insensitive, but **column names are case-sensitive**. Use lowercase for all column names (e.g., `source name`, `characteristics[organism]`, `comment[label]`). Incorrect casing like `Source Name` or `Characteristics[organism]` will cause validation failures.
- **Space sensitivity**: The SDRF is sensitive to spaces in column names (`sourcename` ¬ `source name`). Column names must include appropriate spaces (e.g., `source name`, not `sourcename`) but must NOT have a space before the bracket (e.g., `characteristics[organism]`, not `characteristics [organism]`).
- **Column order**: The SDRF columns follows some structure; first the sample metadata columns in Chapter 7; then the data file metadata columns in Chapter 8; followed by the factor values columns in [study-variables].
- **Extension**: The extension of the SDRF file SHOULD be **sdrf.tsv (preferred)** or .txt.

5.3. Reserved words

There are general scenarios where cell values cannot be provided with actual data. The following reserved words MUST be used in these cases. Reserved words MUST be all lowercase (e.g., `not available`, NOT `Not Available` or `Not available`):

- **not available**: In some cases, the column is mandatory in the format, but for some samples the corresponding value is unknown or could not be determined. In those cases, users SHOULD use **not available**.
- **not applicable**: In some cases, the column is mandatory, but for some samples the corresponding value or concept does not apply. In those cases, users SHOULD use **not applicable**.
- **anonymized**: In some cases, the value exists but has been intentionally redacted for privacy protection (e.g., in clinical studies with de-identified patient data). In those cases, users SHOULD use **anonymized**.
- **pooled**: In some cases, the sample is a pool of multiple samples (e.g., TMT reference channels), and the value cannot be represented as a single value. In those cases, users SHOULD use **pooled**.

Table 1. Reserved words for SDRF cell values

Term	Meaning	Example	Use Case
not available	Value exists but is unknown or could not be determined	characteristics[age] = not available	Patient age was not recorded in the study
not applicable	Value or concept does not apply to this sample	characteristics[age] = not applicable	Synthetic peptide library has no age
anonymized	Value exists but is redacted for privacy protection	characteristics[age] = anonymized	Clinical study with de-identified patient data

Term	Meaning	Example	Use Case
pooled	Value represents a mixture of multiple samples	characteristics[biological replicate] = pooled	TMT reference channel pooled from multiple replicates

5.4. SDRF file-level metadata

Since version 1.1.0, SDRF-Proteomics supports file-level metadata using dedicated columns. These columns provide information about the SDRF file itself, such as the specification version, template(s) used, annotation tool, and validation status. This column-based approach maintains compatibility with spreadsheet applications (Excel, Google Sheets) and existing data processing tools.

The following metadata columns are supported:

Column	Description	Example Value	Requirement	Ontology Term
`comment[sdrf version]`	SDRF-Proteomics specification version used. Should follow semantic versioning format (vMAJOR.MINOR.PATCH)	v1.1.0	RECOMMENDED	PRIDE:0000839
`comment[sdrf template]`	Template name and version used for annotation. Two formats are supported: simple format (`name vX.Y.Z`) or key=value format (`NT=name;VV=vX.Y.Z`). Multiple templates can be specified using multiple columns.	human v1.1.0 or NT=human;VV=v1.1.0	OPTIONAL	PRIDE:0000832
`comment[sdrf annotation tool]`	Software tool, script, or method used to generate or annotate the SDRF file. Two formats are supported: simple format (`name vX.Y.Z`) or key=value format (`NT=name;VV=vX.Y.Z`).	lesSDRF v0.1.0 or NT=lesSDRF;VV=v0.1.0	OPTIONAL	PRIDE:0000840
`comment[sdrf validation hash]`	Cryptographic hash (e.g., SHA-256) generated after successful validation	sha256:abc123…	OPTIONAL	PRIDE:0000834

NOTE When combining multiple templates (e.g., `human` + `ms-proteomics`), use multiple `comment[sdrf template]` columns, one per template. The value in each row should be identical for all samples in the file.

Example of an SDRF file with metadata columns (simplified example showing only select columns; see Chapter 10 for complete required columns):

```
<div class="sdrf-example-table">
<table>
<thead>
```

```
<tr>
<th class="sample-col">source name</th>
<th class="sample-col">characteristics[organism]</th>
<th class="sample-col">characteristics[disease]</th>
<th class="data-col">assay name</th>
<th class="data-col">comment[data file]</th>
<th class="data-col">comment[sdrf version]</th>
<th class="data-col">comment[sdrf template]</th>
<th class="data-col">comment[sdrf template]</th>
<th class="data-col">comment[sdrf annotation tool]</th>
</tr>
</thead>
<tbody>
<tr>
<td class="sample-col">sample_1</td>
<td class="sample-col">homo sapiens</td>
<td class="sample-col">normal</td>
<td class="data-col">run_1</td>
<td class="data-col">sample_1.raw</td>
<td class="data-col">v1.1.0</td>
<td class="data-col">human v1.1.0</td>
<td class="data-col">ms-proteomics v1.1.0</td>
<td class="data-col">lesSDRF v0.1.0</td>
</tr>
<tr>
<td class="sample-col">sample_2</td>
<td class="sample-col">homo sapiens</td>
<td class="sample-col">breast cancer</td>
<td class="data-col">run_2</td>
<td class="data-col">sample_2.raw</td>
<td class="data-col">v1.1.0</td>
<td class="data-col">human v1.1.0</td>
<td class="data-col">ms-proteomics v1.1.0</td>
<td class="data-col">lesSDRF v0.1.0</td>
</tr>
</tbody>
</table>
<div class="sdrf-legend">
<span class="legend-item"><span class="legend-color sample-bg"></span> Sample
metadata</span>
<span class="legend-item"><span class="legend-color data-bg"></span> Data file
metadata</span>
</div>
</div>
```

5.5. Table Column headers

Depending on each section the column headers (property names) will be prefixed with the following prefixes:

- `characteristics`: Sample metadata (e.g. *characteristics[organism]*)
- `comment`: Data file metadata (e.g. *comment[data file]*)
- `factor value`: Factor values properties (e.g. *factor value[disease]*)

Each property name MUST be a valid ontology term or a valid controlled vocabulary term. Each section will have some specific order for column headers.

NOTE A list of all controlled vocabularies and ontologies supported are in the Chapter 12 section. On each section we also provide a list of properties that are supported.

5.6. Table Cell values

The value for each property, (e.g. characteristics, comment, factor value) corresponding to each sample or data file can be represented in multiple ways.

- **Free Text (Human readable)**: In the free text representation, the value is provided as text without Ontology support (e.g. colon or providing accession numbers). This is only RECOMMENDED when the text inserted in the table is the exact name of an ontology/CV term in EFO. If the term is not in EFO, other ontologies can be used.

source name	characteristics[organism]
sample 1	homo sapiens
sample 2	homo sapiens

- **Ontology url (Computer readable)**: Users can provide the corresponding URI (Uniform Resource Identifier) of the ontology/CV term as a value. This is recommended for enriched files where the user does not want to use intermediate tools to map from free text to ontology/CV terms.
- **Key=value representation (Human and Computer readable)**: The current representation aims to provide a mechanism to represent the complete information of the ontology/CV term including Accession, Name and other additional properties. In the key=value pair representation, the Value of the property is represented as an Object with multiple properties, where the key is one of the properties of the object and the value is the corresponding value for the particular key. The key order MUST be `NT` (name) first, followed by `AC` (accession), then any additional keys. An example of key value pairs is post-translational modification (see Protein Modifications):

```
NT=Glu->pyro-Glu;AC=Unimod:27;MT=fixed;PP=Anywhere;TA=E
```

NOTE Beyond these three representations, SDRF columns may accept additional structured value types such as numbers with units (`10 ppm`), accession identifiers (`SAMN12345678`), ISO 8601 dates, semantic versions, and more. Each column's YAML template definition declares exactly which value types and formats are accepted. For the complete reference of all value types, parsing rules, and their formal patterns, see Value Types Reference in the Templates Guide.

Chapter 6. Validating SDRF Files

The official validator for SDRF-Proteomics files is **sdrf-pipelines**, a Python tool that checks your SDRF file for errors and compliance with the specification.

Installation:

```
pip install sdrf-pipelines
```

Basic Validation:

```
# Validate an SDRF file
parse_sdrf validate-sdrf --sdrf_file your_file.sdrf.tsv

# Validate with a specific template
parse_sdrf validate-sdrf --sdrf_file your_file.sdrf.tsv --template human
```

For more information, visit: sdrf-pipelines on GitHub

Chapter 7. SDRF-Proteomics: Samples metadata

The Sample metadata section provides information about the samples of origin and their characteristics. Each sample contains a *source name* (unique identifier) and a set of *characteristics* columns. The first column of the file should be the *source name* and the following columns should be the characteristics of the sample. For example, for any proteomics experiment (human, vertebrate, cell line), the following characteristics should be provided:

- **source name**: Unique sample name (it can be present multiple times if the same sample is used several times in the same dataset)
- **characteristics[organism]**: The organism of the Sample of origin. Values MUST come from NCBI Taxonomy.
- **characteristics[organism part]**: The main normalized anatomical term for the sample (e.g., liver). Values SHOULD come from UBERON or BTO. Use the clearest ontology-backed anatomy term available for cross-study integration; `characteristics[tissue supergroup]` can be used separately for broader organ/system grouping and `characteristics[sampling site]` for finer provenance. `characteristics[sampling site]` may be equal to `characteristics[organism part]` when no distinction exists.
- **characteristics[disease]**: The disease under study in the Sample. Values SHOULD come from MONDO, EFO, or DOID. For healthy/control samples, use `normal` (PATO:0000461) - see Disease Annotation Guidelines.
- **characteristics[cell type]**: A cell type is a distinct morphological or functional form of cell (e.g., epithelial, glial). Values SHOULD come from Cell Ontology (CL), BTO, or Cell Line Ontology (CLO).

Example:

source name	characteristics[organism]	characteristics[organism part]	characteristics[disease]	characteristics[cell type]
sample_treat	homo sapiens	liver	liver cancer	not available
sample_control	homo sapiens	liver	liver cancer	not available

NOTE

- Additional characteristics can be added per experiment type - see SDRF-Proteomics templates for required properties.
- Column headers SHOULD use EFO ontology terms (e.g., `characteristics[organism]`) - see Disease Annotation Guidelines.
- Multiple columns with the same `characteristics` term are allowed (see Section 9.1), but RECOMMENDED to use more specific terms (e.g., "immunophenotype" instead of duplicate "phenotype").

7.1. BioSamples database integration

Use the OPTIONAL `characteristics[biosample accession number]` column to link samples to BioSamples [5], enabling cross-database integration with genomics and transcriptomics data. Formats:

`SAMN*` (NCBI) or `SAMEA*` (EBI).

7.2. Encoding sample technical and biological replicates

SDRF-Proteomics uses two REQUIRED columns to track replicates [4]:

- **characteristics[biological replicate]**: Independent biological samples. Numbering restarts per experimental condition (factor value group).
- **comment[technical replicate]**: Repeated measurements of the same sample (e.g., multiple injections)

When no replicates are performed, set both columns to `1`. For pooled samples, use `pooled` for biological replicate.

source name	characteristics[biological replicate]	comment[fraction identifier]	comment[technical replicate]	comment[data file]
patient_001	1	1	1	P001_F1_TR1.raw
patient_001	1	1	2	P001_F1_TR2.raw
patient_002	2	1	1	P002_F1_TR1.raw
patient_002	2	1	2	P002_F1_TR2.raw

7.3. Pooled samples

When multiple samples are pooled into one (e.g., TMT/iTRAQ reference channels for normalization), use the *characteristics[pooled sample]* column to indicate pooling status. Allowed values:

- **not pooled**: Regular individual samples
- **pooled**: Sample is pooled but individual sources are unknown
- **SN=sample1;SN=sample2;…**: Lists source names of pooled samples when known

Example:

source name	characteristics[pooled sample]	characteristics[organism]	characteristics[age]	comment[label]	comment[data file]
sample_1	not pooled	homo sapiens	45Y	TMT126	file01.raw
sample_2	not pooled	homo sapiens	52Y	TMT127N	file01.raw
pooled_ref	SN=sample_1;SN=sample_2	homo sapiens	pooled	TMT131C	file01.raw

TIP For pooled samples, use `pooled` for individual-specific fields (biological replicate, age, sex) to indicate a mixture rather than a single sample.

7.4. Sample Metadata Guidelines

For detailed guidance on annotating sample metadata, refer to the following conventions documents:

- Sample Metadata Guidelines - Detailed guidelines for age, sex, disease, organism part, cell type, developmental stage, spiked-in samples, and other sample characteristics
- Human Sample Metadata Guidelines - Human-specific metadata including disease staging, treatment history, demographics, and lifestyle factors

Chapter 8. SDRF-Proteomics: data files metadata

The connection between samples and data files is done using properties annotated with the `comment` prefix. All properties referring to a data file (e.g., MS run file) are annotated with the category `comment`. This differentiates data file properties from sample properties (characteristics).

8.1. CV Term Format for Data File Metadata

For data file metadata (`comment` columns) that reference ontology terms, use the structured format: `NT={term name};AC={accession}`

Examples: `NT=HCD;AC=PRIDE:0000590`, `NT=Orbitrap;AC=MS:1000484`

This format enables automated validation and software extraction from raw files. Sample metadata (characteristics) can use simple term names since they are typically human-annotated.

The following properties MUST be provided for each data file in **mass spectrometry-based proteomics** experiments. For **affinity-based proteomics** (Olink, SomaScan), see the Affinity-Proteomics template for different required columns.

Column	Requirement	Description	Ontology
`assay name`	REQUIRED	Unique identifier for an MS run/data file	Free text
`technology type`	REQUIRED	Technology used to capture the data	Fixed values
`comment[proteomics data acquisition method]`	REQUIRED	DDA, DIA, PRM, SRM	PRIDE:0000659
`comment[label]`	REQUIRED	Label applied to sample (or "label free sample")	PRIDE - Labels
`comment[instrument]`	REQUIRED	Mass spectrometer model	PSI-MS - Instruments
`comment[cleavage agent details]`	REQUIRED	Enzyme information (use "not applicable" for top-down/undigested samples)	PSI-MS - Cleavage agents
`comment[fraction identifier]`	REQUIRED	Fraction number (1 if not fractionated)	Integer
`comment[technical replicate]`	REQUIRED	Technical replicate number (1 if none)	Integer
`comment[data file]`	REQUIRED	Name of the raw file (primary/canonical file when the vendor format ships multiple files - see Section 8.2)	Free text

Example:

source name	assay name	technology type	comment[proteomics data acquisition method]	comment[label]	comment[instrument]	comment[data file]
sample_1	sample1_run1	proteomic profiling by mass spectrometry	data-dependent acquisition	label free sample	Q Exactive HF	sample1.raw

8.2. Vendor formats that ship multiple files (e.g. AB Sciex .wiff + .wiff.scan)

Some vendor formats persist a single MS run as **two or more files on disk** that are only useful together. The most common case is AB Sciex, where every acquisition produces a `.wiff` file paired with a `.wiff.scan` (and sometimes `.wiff2`) sidecar; tools such as ProteoWizard/Skyline cannot open one without the other. Bruker `.d` directories, in contrast, are a single logical entity (a folder) and do **not** require this pattern - reference the `.d` folder name directly in `comment[data file]`.

To represent these multi-file formats while keeping one row per MS run, SDRF-Proteomics treats one file as the **primary/canonical** file and the remaining files as **auxiliary** files described by sidecar columns:

Column	Cardinality	Description
`comment[data file]`	1	Name of the primary/canonical file for the MS run (for AB Sciex, the `.wiff` file).
`comment[file uri]`	1	URI to retrieve the primary file. The trailing path segment MUST equal `comment[data file]`.
`comment[associated data file]`	*	Name of an auxiliary file that must be downloaded alongside the primary file (for AB Sciex, the `.wiff.scan` file). Repeat the column when more than one auxiliary file is required.
`comment[associated file uri]`	*	URI to retrieve the auxiliary file. The trailing path segment MUST equal the corresponding `comment[associated data file]`. Repeat in the same order as `comment[associated data file]`.

NOTE When multiple `comment[associated data file]` / `comment[associated file uri]` columns are present, consumers MUST pair them positionally (first associated file with first associated URI, second with second, etc.).

Example - AB Sciex (.wiff + .wiff.scan):

source name	assay name	comment[instrument]	comment[data file]	comment[file uri]	comment[associated data file]	comment[associated file uri]
sample_1	sample1_run1	TripleTOF 6600	OA_3.wiff	https://ftp.pride.ebi.ac.uk/pride/data/archive/2026/03/PXD073289/OA_3.wiff	OA_3.wiff.scan	https://ftp.pride.ebi.ac.uk/pride/data/archive/2026/03/PXD073289/OA_3.wiff.scan

TIP

Once repositories standardise on bundling related files into a single `.zip` or `.tar.gz` (as already done for some Bruker timsTOF deposits), `comment[data file]` and `comment[file uri]` can reference the archive directly and `comment[associated data file]` is no longer needed. See the PRIDE submission formats guidelines for the direction of travel.

If a repository already distributes a single canonical archive for one acquisition today (for example a `.raw.zip` bundle in some metabolomics deposits), that archive can already be recorded directly in `comment[data file]` and `comment[file uri]`.

8.3. Sample Preparation and Fragmentation (MS-based only)

NOTE

This section applies to **mass spectrometry-based proteomics** experiments only. For affinity-based proteomics, these properties do not apply.

For detailed documentation of sample preparation and MS/MS fragmentation properties, see the MS-Proteomics Template:

- **Sample preparation**: depletion, reduction reagent, alkylation reagent
- **Fractionation**: fractionation method (used with `comment[fraction identifier]`)
- **Fragmentation**: collision energy, dissociation method

NOTE

For HCD (Higher-energy C-trap Dissociation), the canonical accession is MS:1000422 - beam-type collision-induced dissociation. Use `NT=beam-type collision-induced dissociation;AC=MS:1000422` or the short label `HCD`. Do not use PRIDE:0000590 or MS:1002481.

8.4. Proteomics data acquisition method

Proteomics data acquisition method can happen in multiple ways: Data Dependent Acquisition (DDA), Data Independent Acquisition (DIA), and targeted approaches. The SDRF-Proteomics file format REQUIRES capturing the method used for the data acquisition in the *comment[proteomics data acquisition method]* column. The values MUST be children of the PRIDE ontology term proteomics data acquisition method (PRIDE:0000659). The following values are commonly used:

- data-dependent acquisition

- data-independent acquisition
  - diaPASEF
  - SWATH MS
- parallel reaction monitoring
- selected reaction monitoring

IMPORTANT	The *comment[proteomics data acquisition method]* column is REQUIRED for all mass spectrometry-based SDRF files. This field must be explicitly specified and cannot be omitted or assumed.

You can find an example of a DIA experiment in the following link: DIA example

TIP	For DIA experiments, additional properties like MS1 scan range can be captured. See DIA Scan Window Limits in the DIA-Acquisition Template.

8.5. MS-Proteomics Template

For detailed guidance on data file metadata, refer to the conventions document:

- MS-Proteomics Template - Detailed guidelines for labels, instruments, modifications, cleavage agents, mass tolerances, RAW file URIs, and other data file properties

Chapter 9. Additional SDRF Rules

9.1. Column Cardinality

Some columns can appear multiple times for the same sample. The cardinality rules are:

- **Single (1)**: Column appears exactly once per sample (e.g., `characteristics[biological replicate]`)
- **Multiple (*)**: Column can appear multiple times (e.g., `comment[modification parameters]` can specify multiple post-translational modifications)

Example of multiple `comment[modification parameters]` columns:

source name	characteristics[…]	comment[modification parameters]	comment[modification parameters]	…
sample-1	…	NT=Carbamidomethyl;AC=UNIMOD:4;TA=C;MT=fixed;PP=Anywhere	NT=Oxidation;AC=UNIMOD:35;TA=M;MT=variable;PP=Anywhere	…

9.2. Row Uniqueness Requirements

Uniqueness constraints ensure data integrity:

- **MUST be unique** (error): `source name` + `assay name` + `comment[label]`
- **SHOULD be unique** (warning): `source name` + `assay name`
- **Assay name**: Each data file MUST have a unique `assay name`

> **NOTE** For multiplexed experiments (TMT, iTRAQ), multiple rows share the same `assay name` since samples are in one MS run. The `comment[label]` distinguishes samples within the run.

Chapter 10. Templates

A **template** is a predefined set of metadata columns that ensures consistent annotation for specific experiment types. Templates define REQUIRED, RECOMMENDED, and OPTIONAL columns to make datasets FAIR-compliant.

10.1. Template Architecture

Templates follow a layered hierarchy:

Layer	Templates	Description
TECHNOLOGY (required)	ms-proteomics, affinity-proteomics, ms-metabolomics	Minimum valid SDRF - choose one
SAMPLE (recommended)	human, vertebrates, invertebrates, plants, clinical-metadata, oncology-metadata	Organism-specific and clinical metadata
EXPERIMENT (optional)	cell-lines, crosslinking, dia-acquisition, single-cell, immunopeptidomics, metaproteomics, lc-ms-metabolomics, gc-ms-metabolomics	Methodology-specific columns

Child templates inherit all columns from parents and may add new columns or strengthen requirements (e.g., `optional` ¬ `required`).

10.2. Template Combination Rules

Some layers enforce **mutually exclusive** choices, while others allow combining multiple templates:

Layer	Templates	Rule
TECHNOLOGY	`ms-proteomics` vs `affinity-proteomics` vs `ms-metabolomics`	Mutually exclusive — choose one (REQUIRED)
SAMPLE	`human` vs `vertebrates` vs `invertebrates` vs `plants`	Mutually exclusive — choose one based on organism (RECOMMENDED)
EXPERIMENT (MS)	`dia-acquisition`, `single-cell`, `crosslinking`, `immunopeptidomics`, `metaproteomics`	Can be combined (e.g., `dia-acquisition` + `single-cell`)
EXPERIMENT (metabolomics)	`lc-ms-metabolomics` vs `gc-ms-metabolomics`	Mutually exclusive — choose one when using `ms-metabolomics` (OPTIONAL)

Templates from different layers can be freely combined. Common valid combinations:

- `ms-proteomics` + `human` (human DDA proteomics)
- `ms-proteomics` + `human` + `dia-acquisition` (human DIA proteomics)
- `ms-proteomics` + `human` + `immunopeptidomics` (human immunopeptidomics)
- `ms-proteomics` + `vertebrates` + `cell-lines` (mouse cell line proteomics)
- `ms-proteomics` + `human` + `crosslinking` (human crosslinking MS)
- `affinity-proteomics` + `human` (human affinity proteomics, including Olink or SomaScan)
- `ms-metabolomics` + `human` + `lc-ms-metabolomics` (human LC-MS metabolomics)
- `ms-metabolomics` + `plants` + `gc-ms-metabolomics` (plant GC-MS metabolomics)
- `ms-proteomics` + `metaproteomics` (environmental metaproteomics)
- `ms-proteomics` + `human-gut` (human gut microbiome metaproteomics)
- `ms-proteomics` + `human` + `single-cell` (human single-cell proteomics)

10.3. Specifying Templates in SDRF Files

Declare templates using `comment[sdrf template]` columns. Only list leaf templates (parents are implied). When using multiple templates, add multiple columns with the same name. Two formats are supported:

- Simple format (preferred): `template_name vX.Y.Z`
- Key=value format: `NT=template_name;VV=vX.Y.Z`

```
source name ... comment[sdrf template]  comment[sdrf template]
sample_1    ... human v1.1.0    crosslinking v1.0.0
```

Common examples:

Experiment Type	Template Columns
Human MS proteomics	`comment[sdrf template]` = `human v1.1.0`
Mouse MS proteomics	`comment[sdrf template]` = `vertebrates v1.1.0`
Human crosslinking	Two columns: `human v1.1.0` + `crosslinking v1.0.0`
Human affinity proteomics	Two columns: `human v1.1.0` + `affinity-proteomics v1.0.0`
Human LC-MS metabolomics	Two columns: `human v1.1.0` + `lc-ms-metabolomics v1.0.0-dev`

10.4. Available Templates

Sample templates (organism-specific):

Template	Use For	Key Columns
Human	Human clinical samples	disease, age, sex, ancestry
Vertebrates	Mouse, rat, zebrafish	disease, developmental stage, strain
Invertebrates	Drosophila, C. elegans	disease, developmental stage, genotype
Plants	Arabidopsis, crops	disease, developmental stage, growth conditions

Experiment-type templates:

- Affinity Proteomics - Olink and SomaScan-style platform metadata
- DIA Acquisition - scan windows, isolation width
- Cell Lines - Cellosaurus integration
- Single-Cell - cell isolation, carrier proteome
- Immunopeptidomics - MHC protein complex, MHC typing
- Crosslinking MS - crosslinker reagents
- LC-MS Metabolomics - chromatography, ion source, polarity, scan range
- GC-MS Metabolomics - derivatization, GC column, carrier gas, oven program
- Metaproteomics - environmental sample type

Download templates from the templates folder.

10.5. Extending Templates

You can add custom columns beyond template requirements for study-specific metadata. Rules:

- Use `characteristics[…]` for sample metadata, `comment[…]` for technical metadata
- Column names MUST be valid ontology terms (search OLS)
- Use controlled vocabularies for values when available

See Additional Sample-Related Columns and SDRF Terms Reference for commonly used columns.

10.6. Contributing New Templates

To propose a new template, open an issue on GitHub and submit a pull request.

Chapter 11. Factor Values (Study Variables)

Factor values identify the experimental variables being studied - the conditions you want to compare in your analysis. They highlight which sample characteristics are the focus of your experiment.

11.1. Column Format

`factor value[{variable name}]`

11.2. When to Use Factor Values

Use factor values to indicate:

- The primary variable(s) under investigation
- Conditions being compared (e.g., disease vs. normal, treated vs. untreated)
- Variables that define experimental groups

> **NOTE** Use `normal` (not "control") in the disease field for healthy samples. "Control" is an experimental design concept, not a disease state. See Disease Annotation Guidelines for details.

11.3. Rules

- Factor value columns SHOULD appear after all characteristics and comment columns
- Multiple factor values can be used when studying multiple variables
- The value in a factor value column typically mirrors a characteristics column value

11.4. Example

In an experiment comparing tumor vs. normal tissue across different cancer stages:

source name	...	characteristics[disease]	characteristics[disease staging]	...	factor value[disease]	factor value[disease staging]
tumor_sample_1	…	breast carcinoma	stage II	…	breast carcinoma	stage II
normal_sample_1	…	normal	not applicable	…	normal	not applicable
tumor_sample_2	…	breast carcinoma	stage III	…	breast carcinoma	stage III

In this example, both `disease` and `disease staging` are factor values because the experiment aims to compare expression differences between disease states and across cancer stages.

Chapter 12. Ontologies and Controlled Vocabularies

SDRF-Proteomics uses ontologies and controlled vocabularies (CVs) to standardize metadata values. The following ontologies are supported:

Category	Ontology/CV	Description	Notes
General Purpose			
General	Experimental Factor Ontology (EFO)	General experimental metadata	
General	PATO	Phenotype and Trait Ontology	
General	NCI Thesaurus (NCIT)	Biomedical terminology	
General	PRIDE Controlled Vocabulary	Proteomics-specific terms	
Organism and Taxonomy			
Taxonomy	NCBI Taxonomy (NCBITaxon)	Organism classification	
Anatomy and Cell Types			
Anatomy	UBERON	Cross-species anatomy ontology	
Cell Type	Cell Ontology (CL)	Cell type classification	
Anatomy	BRENDA Tissue Ontology (BTO)	Tissues and cell lines	
Anatomy	Plant Ontology (PO)	Plant anatomy and development	For plant samples
Anatomy	FlyBase Anatomy (FBbt)	Drosophila anatomy	For Drosophila samples
Anatomy	WormBase Anatomy (WBbt)	C. elegans anatomy	For C. elegans samples
Anatomy	Zebrafish Anatomy (ZFA)	Zebrafish anatomy and development	For zebrafish samples
Disease (see Disease Annotation Guidelines)			
Disease	Mondo Disease Ontology (MONDO)	Unified disease ontology	RECOMMENDED
Disease	Experimental Factor Ontology (EFO)	Disease terms from EFO	
Healthy samples	Phenotype And Trait Ontology (PATO)	Use `normal` (PATO:0000461) for healthy samples	
Cell Lines			
Cell Lines	Cellosaurus	Cell line knowledge resource	RECOMMENDED

Category	Ontology/CV	Description	Notes
Cell Lines	Cell Line Ontology (CLO)	Cell line ontology	
Mass Spectrometry and Proteomics			
MS/Proteomics	PSI Mass Spectrometry CV (PSI-MS)	Instruments, methods, parameters	
Modifications	Unimod	Protein modifications database	
Modifications	PSI-MOD CV	Protein modifications ontology	
Other			
Chemistry	ChEBI	Chemical Entities of Biological Interest	
Environment	Environment Ontology (ENVO)	Environmental sample classification	For metaproteomics
Ancestry	Human Ancestry Ontology (HANCESTRO)	Human ancestry categories	For human samples

Chapter 13. Examples of Annotated Datasets

The following table provides links to example SDRF files for different experiment types. Click "View in Explorer" to open the SDRF file in the interactive viewer.

Experiment Type	Dataset	Description	View	Source
Label-free	PXD008934	Human proteome label-free quantification	View in Explorer	GitHub
TMT	PXD017710	TMT-labeled quantitative proteomics	View in Explorer	GitHub
SILAC	PXD000612	SILAC-based quantification	View in Explorer	GitHub
DIA	PXD018830	data-independent acquisition	View in Explorer	GitHub
Phosphoproteomics	PXD000759	PTM enrichment study	View in Explorer	GitHub
Cell lines	PXD001819	Cell line proteomics	View in Explorer	GitHub

TIP Use the SDRF Explorer to browse all {total_datasets}+ annotated datasets with filtering, statistics, and interactive viewing.

A comprehensive collection of annotated projects is available at: Annotated Projects Repository

Chapter 14. Template Definitions

This section provides the column definitions for each SDRF-Proteomics template. Each template shows only its **own** columns (not inherited ones). See the "Extends" field to identify which parent template's columns are also included.

14.1. base

Version: 1.1.0 | **Layer:** internal | **Extends:** none | **Usable alone:** No

Base SDRF template with infrastructure columns (identifiers, data files, versioning) inherited by all proteomics templates. This is a construction artifact and cannot be used directly.

Column Name	Req.	Description	Validators	Examples
`source name`	required	Unique identifier for the biological sample		
`assay name`	required	Unique identifier for the data acquisition run		
`technology type`	required	Type of technology used	single value only; values: proteomic profiling by mass spectrometry, protein expression profiling by antibody array, protein expression profiling by aptamer array	
`comment[technical replicate]`	required	Identifier for the technical replicate (integer starting from 1)		
`comment[data file]`	required	Name of the raw data file		
`comment[sdrf version]`	recommended	Version of the SDRF-Proteomics specification used to annotate this file	semver	v1.1.0, v2.0.0-dev
`comment[sdrf template]`	optional	Template name and version used for annotation. Two formats are supported - key=value format (NT=template_name;VV=vX.Y.Z) or simple format (template_name vX.Y.Z). Multiple templates can be specified using multiple columns.	pattern: Template can be specified as 'NT=name;VV=vX.Y.Z' or 'name vX.Y.Z'	NT=human;VV=v1.1.0, human v1.1.0, NT=ms-proteomics;VV=v1.1.0, ms-proteomics v1.1.0

Column Name	Req.	Description	Validators	Examples
`comment[sdrf annotation tool]`	optional	Software tool or method used to generate or annotate the SDRF file. Two formats are supported - key=value format (NT=tool_name;VV=vX.Y.Z) or simple format (tool_name vX.Y.Z).	pattern: Annotation tool can be specified as 'NT=name;VV=vX.Y.Z' or 'name vX.Y.Z' or 'manual curation'	NT=lesSDRF;VV=v0.1.0, lesSDRF v0.1.0, NT=sdrf-pipelines;VV=v1.0.0, sdrf-pipelines v1.0.0, …
`comment[sdrf validation hash]`	optional	Hash value for SDRF validation integrity checking	pattern: Validation hash string	

14.2. sample-metadata

Version: 1.0.0 | **Layer:** internal | **Extends:** base | **Usable alone:** No

SDRF template with shared sample metadata columns (organism, tissue, disease). This is an internal construction layer inherited by technology and organism templates - not used directly.

Column Name	Req.	Description	Validators	Examples
`characteristics[organism]`	required	Species of the sample using NCBI Taxonomy	ontology: ncbitaxon	homo sapiens, mus musculus, rattus norvegicus, saccharomyces cerevisiae
`characteristics[organism part]`	required	Main normalized anatomical term for the sample	ontology: uberon, bto	liver, brain, heart, blood
`characteristics[tissue supergroup]`	optional	Broader anatomical grouping or system-level bucket for the sample, used alongside the organism part for higher-level grouping	ontology: uberon, bto	heart, liver, brain, gastrointestinal tract
`characteristics[cell type]`	recommended	Cell type of the sample	ontology: cl, bto, clo	hepatocyte, neuron, fibroblast, T cell
`characteristics[biological replicate]`	required	Identifier for the biological replicate (integer starting from 1, or 'pooled' for pooled samples)	pattern: Biological replicate should be an integer or 'pooled' for pooled reference samples	1, 2, pooled

Column Name	Req.	Description	Validators	Examples
`characteristics[pooled sample]`	optional	Whether the sample is a pooled sample combining material from multiple biological sources. Use 'not pooled' for individual samples, 'pooled' when sources are unknown, or 'SN=sample1;SN=sample2' to list source names.	values: not pooled, pooled; pattern: Use 'not pooled', 'pooled', or list sample IDs with SN= prefix	SN=sample1;SN=sample2
`characteristics[sample type]`	optional	Classification of the sample role in the experiment. Distinguishes experimental samples from controls, references, and other roles in multiplexed or plate-based experiments.	ontology: pride	single cell, reference, bridge, carrier, …
`characteristics[disease]`	recommended	Disease state of the sample	ontology: mondo, efo, doid, ncit, pato	normal, breast cancer, infection, metabolic disease
`characteristics[material type]`	optional	Type of biological material being analyzed	values: tissue, cell, cell line, organism part, …	
`characteristics[tissue mass]`	optional	Mass of tissue used for extraction	number with unit (mg, g, ug)	50 mg, 1 g, 500 ug
`characteristics[biosample accession number]`	optional	BioSample accession number for the sample (e.g., SAMN or SAMEA identifiers)	accession: biosample	SAMN12345678, SAMEA12345678, SAMD1234567
`characteristics[sampling time]`	optional	Time at which the sample was collected (for longitudinal or time-course studies)	number with unit (hour, day, minute, week, month, year)	0 hour, 24 hour, 7 day, 3 month
`characteristics[treatment]`	optional	Treatment or perturbation applied to the sample (drug, stimulus, environmental stress)	ontology: ncit, efo	untreated, LPS stimulation, doxorubicin treatment, drought stress, …
`characteristics[synthetic peptide]`	optional	Whether the sample is a synthetic peptide library or biological material	values: synthetic, not synthetic	

Column Name	Req.	Description	Validators	Examples
`characteristics[spiked compound]`	optional	Spiked-in compound details using key-value format (CT=compound type, QY=quantity, PS=peptide sequence, AC=UniProt accession, CN=compound name, CV=vendor)	pattern: Key-value format for spiked compound details (CT=type, SP=species, QY=quantity, PS=sequence, AC=accession, CN=name, CV=vendor)	CT=peptide;PS=PEPTIDESEQ;QY=10 fmol, CT=protein;AC=A9WZ33;QY=20 nmol, CT=protein;SP=Homo sapiens;QY=1 pmol;AC=P37840, CT=mixture;CN=iRT mixture;CV=Biognosys;QY=1 pmol
`characteristics[enrichment process]`	optional	Enrichment strategy applied to the sample (e.g., phosphopeptide enrichment, crosslinked peptide enrichment, glycopeptide enrichment)	ontology: pride, efo	enrichment of cross-linked peptides, enrichment of phosphorylated protein, enrichment of glycopeptides, enrichment of ubiquitinated proteins

14.3. ms-proteomics

Version: 1.1.0 | **Layer:** technology | **Extends:** sample-metadata | **Usable alone:** Yes

Base SDRF template for mass spectrometry-based proteomics. This is the minimum valid template for any MS experiment.

Column Name	Req.	Description	Validators	Examples
`comment[proteomics data acquisition method]`	required	Mass spectrometry acquisition method	ontology: pride	data-dependent acquisition, data-independent acquisition, parallel reaction monitoring, selected reaction monitoring
`comment[instrument]`	required	Mass spectrometer instrument used	ontology: ms, pride	LTQ Orbitrap, Q Exactive, Orbitrap Fusion Lumos, timsTOF Pro
`comment[cleavage agent details]`	required	Enzyme or chemical used for protein digestion	ontology: ms	NT=Trypsin;AC=MS:1001251, NT=Lys-C;AC=MS:1001309, NT=Chymotrypsin;AC=MS:1001306

Column Name	Req.	Description	Validators	Examples
comment[label]	required	Labeling strategy used for quantification	ontology: pride	label free sample, SILAC light, SILAC heavy, TMT126, …
comment[fraction identifier]	required	Fraction number for fractionated samples (integer, use 1 for non-fractionated). In MS proteomics, this identifies the chromatographic or electrophoretic fraction (e.g., SCX, hpHRP, SEC fractions). Each fraction maps to one data file.		
comment[dissociation method]	recommended	Fragmentation method used in MS/MS	ontology: ms, pride	HCD, CID, ETD, EThcD
comment[fractionation method]	optional	Peptide fractionation method used before MS analysis	ontology: pride	High-pH reversed-phase chromatography (hpHRP), Strong cation-exchange chromatography (SCX), Strong anion-exchange chromatography (SAX), Size-exclusion chromatography (SEC)
comment[collision energy]	optional	Collision energy used for fragmentation	pattern: Collision energy format: {value} {unit} where unit is NCE or eV. For multiple values, use semicolon-separated entries.	30 NCE, 30% NCE, 27 eV, 25 NCE;27 NCE;30 NCE
comment[precursor mass tolerance]	recommended	Precursor mass tolerance for database search	number with unit (ppm, Da, mmu)	10 ppm, 20 ppm, 0.5 Da, 20 mmu
comment[fragment mass tolerance]	recommended	Fragment mass tolerance for database search	number with unit (ppm, Da, mmu)	0.02 Da, 20 ppm, 50 mmu
comment[reduction reagent]	optional	Chemical reagent used for disulfide bond reduction	ontology: pride, ms	dithiothreitol, tris(2-carboxyethyl)phosphine
comment[alkylation reagent]	optional	Chemical reagent used for cysteine alkylation	ontology: pride, ms	iodoacetamide, chloroacetamide

Column Name	Req.	Description	Validators	Examples
`characteristics[depletion]`	optional	Whether abundant protein depletion was performed	values: no depletion, depletion	
`comment[modification parameters]`	recommended	Post-translational modifications searched	ontology: unimod, mod	NT=Oxidation;MT=Variable;TA=M;AC=Unimod:35, NT=Carbamidomethyl;TA=C;MT=fixed;AC=UNIMOD:4
`comment[ms2 mass analyzer]`	optional	Mass analyzer used for MS2 acquisition	ontology: ms	orbitrap, ion trap, TOF
`comment[sample preparation batch]`	optional	Batch identifier for sample preparation (plate, chip, processing batch). Useful for batch effect correction in multi-batch experiments.	pattern: Sample preparation batch identifier	plate1, batch_20220601, prep_A
`comment[lc batch]`	optional	Liquid chromatography batch identifier for batch effect tracking (e.g., column changes, LC system swaps)	pattern: LC batch identifier	LC1, column_A
`comment[acquisition date]`	optional	Date of MS data acquisition (ISO 8601 format recommended). Useful for tracking instrument drift and batch effects.	pattern: Acquisition date/time	2022-06-01, 2022-06-01T18:28:37
`comment[ms min mz]`	optional	MS method-defined minimum precursor (MS1) m/z setting used to acquire the data	m/z value	100m/z, 200m/z, 350.5m/z
`comment[ms max mz]`	optional	MS method-defined maximum precursor (MS1) m/z setting used to acquire the data	m/z value	1200m/z, 1600m/z, 2000m/z
`comment[ms min charge]`	optional	MS method-defined minimum precursor charge state setting used to acquire the data	pattern: Integer charge state	1, 2
`comment[ms max charge]`	optional	MS method-defined maximum precursor charge state setting used to acquire the data	pattern: Integer charge state	6, 7, 8
`comment[ms min rt]`	optional	LC method-defined minimum retention time setting used to acquire the data (in minutes)	pattern: Numeric retention time in minutes	0, 5, 10.5

Column Name	Req.	Description	Validators	Examples
comment[ms max rt]	optional	LC method-defined maximum retention time setting used to acquire the data (in minutes)	pattern: Numeric retention time in minutes	60, 90, 120
comment[ms min im]	optional	MS method-defined minimum ion mobility setting used to acquire the data (1/K0 or Vs/cm2)	pattern: Numeric ion mobility value	0.6, 0.7
comment[ms max im]	optional	MS method-defined maximum ion mobility setting used to acquire the data (1/K0 or Vs/cm2)	pattern: Numeric ion mobility value	1.3, 1.4, 1.6
comment[ms2 min mz]	optional	MS method-defined minimum product ion (MS2) m/z setting used to acquire the data	m/z value	100m/z, 200m/z
comment[ms2 max mz]	optional	MS method-defined maximum product ion (MS2) m/z setting used to acquire the data	m/z value	1800m/z, 2000m/z
comment[ms3 min mz]	optional	MS method-defined minimum product ion (MS3) m/z setting used to acquire the data	m/z value	100m/z, 200m/z
comment[ms3 max mz]	optional	MS method-defined maximum product ion (MS3) m/z setting used to acquire the data	m/z value	1500m/z, 2000m/z
comment[ms1 scan range]	optional	m/z scan range for MS1 spectra as an interval. Alternative to separate ms min mz / ms max mz columns	m/z range interval	400m/z-1200m/z, 350m/z-1600m/z
comment[ms2 scan range]	optional	m/z scan range for MS2 spectra as an interval. Alternative to separate ms2 min mz / ms2 max mz columns	m/z range interval	100m/z-2000m/z, 200m/z-1800m/z
comment[ms3 scan range]	optional	m/z scan range for MS3 spectra as an interval. Alternative to separate ms3 min mz / ms3 max mz columns	m/z range interval	100m/z-1500m/z, 200m/z-2000m/z
comment[elution conditions]	optional	Conditions used for peptide/protein elution	pattern: Free-text elution conditions	0.1% TFA in water, 80% acetonitrile, gradient 5-35% ACN in 60 min

14.4. affinity-proteomics

Version: 1.0.0 | **Layer:** technology | **Extends:** sample-metadata | **Usable alone:** Yes

SDRF template for affinity-based proteomics experiments (Olink, SomaScan). This is the base template for all affinity proteomics experiments.

Column Name	Req.	Description	Validators	Examples
`characteristics[sample type]`	required	Role of the sample in the assay. Distinguishes study samples from controls/reference materials for assay interpretation and quality control	ontology: pride	study sample, negative control, positive control, calibrator, …
`comment[platform]`	required	Affinity proteomics platform used (e.g. Olink Explore HT, SomaScan Assay 7K)	single value only; ontology: pride	Olink Explore HT, Olink Target 96, SomaScan Assay 11K
`comment[instrument]`	optional	Instrument used for data acquisition (e.g. sequencer, qPCR machine, microarray reader)	ontology: ms, pride	Illumina NovaSeq X, Illumina NextSeq 2000, Agilent SureScan Microarray Scanner
`comment[panel name]`	recommended	Commercial panel/menu identifier within the platform. RECOMMENDED, but MUST be provided when not uniquely inferable from comment[platform].	pattern: Panel/menu identifier	Target 96 Inflammation, Explore 1536, SomaScan 7K, SomaScan 11K
`comment[quantification unit]`	optional	Unit of quantification for the assay (platform-specific)	values: NPX, RFU	
`comment[plate]`	optional	Plate identifier for batch effect analysis	pattern: Plate identifier	1, 2
`comment[normalization method]`	optional	Normalization method applied to quantification values	pattern: Normalization method	plate control normalized, bridge normalized, median normalization, not normalized
`comment[dilution]`	optional	Sample dilution factor used (commonly used for SomaScan workflows)	pattern: Dilution factor; SomaScan commonly uses 0.005%, 0.5%, 20%, 40%	0.005%, 0.5%, 20%, 40%, …
`comment[lot number]`	optional	Reagent lot number for traceability	pattern: Lot number	SS-2023-001, lot_12345

14.5. human

Version: 1.1.0 | **Layer:** sample | **Extends:** sample-metadata | **Usable alone:** No

Human SDRF template with human-specific sample metadata fields. Must be combined with a technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[disease]`	required	*(override: requirement set to required)*		
`characteristics[ancestry category]`	recommended	Ancestry or ethnic background of the donor	ontology: hancestro	European, African, Asian, Hispanic or Latin American
`characteristics[age]`	required	Age of the donor at sample collection	pattern: Age format: 45Y, 6M, 30Y6M (Y>M>W>D order), ranges like 40Y-50Y, or comparison operators like >18Y, >=21Y, <65Y. Use "not available" if unknown, "anonymized" if redacted, or "pooled" for pooled samples.	45Y, 6M, 30Y6M, 30Y6M2W, …
`characteristics[sex]`	required	Biological sex of the donor	values: male, female, intersex	
`characteristics[developmental stage]`	optional	Developmental stage of the donor	ontology: efo	adult, embryonic stage, fetal stage, infant stage
`characteristics[individual]`	recommended	Unique identifier for the donor individual	identifier	patient_001, donor-A1, subject_12, anonymized, …

14.6. vertebrates

Version: 1.1.0 | **Layer:** sample | **Extends:** sample-metadata | **Usable alone:** No

SDRF template for non-human vertebrate samples (mammals, birds, fish, reptiles, amphibians). Must be combined with a technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[disease]`	required	*(override: requirement set to required)*		

Column Name	Req.	Description	Validators	Examples
`characteristics[developmental stage]`	required	Developmental stage of the organism	ontology: efo	adult, embryo, juvenile, larval stage
`characteristics[strain or breed]`	recommended	Strain or breed of the organism	ontology: ncbitaxon	C57BL/6, Sprague-Dawley, BALB/c, Wistar
`characteristics[sex]`	recommended	Biological sex of the organism	values: male, female, hermaphrodite	

14.7. invertebrates

Version: 1.1.0 | **Layer:** sample | **Extends:** sample-metadata | **Usable alone:** No

SDRF template for invertebrate samples (Drosophila, C. elegans, insects, etc.). Must be combined with a technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[disease]`	required	*(override: requirement set to required)*		
`characteristics[developmental stage]`	required	Developmental stage of the organism	ontology: efo	adult stage, larval stage, pupal stage, embryonic stage
`characteristics[strain or breed]`	required	Strain of the organism	ontology: ncbitaxon	Oregon-R, w1118, N2, Canton-S
`characteristics[genotype]`	optional	Genotype of the organism	pattern: Genotype notation following standard conventions	wild type, daf-2(e1370), w[*]; P{GAL4}

14.8. plants

Version: 1.1.0 | **Layer:** sample | **Extends:** sample-metadata | **Usable alone:** No

SDRF template for plant samples (Arabidopsis, crops, etc.). Must be combined with a technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[organism part]`			ontology: uberon, bto, po	flower bud, leaf, root, seed

Column Name	Req.	Description	Validators	Examples
`characteristics[disease]`	required	*(override: requirement set to required)*		
`characteristics[developmental stage]`	required	Developmental stage of the plant	ontology: efo	seedling stage, flowering stage, rosette growth stage, senescent stage
`characteristics[strain or breed]`	recommended	Cultivar, ecotype, or accession of the plant	pattern: Plant cultivar or ecotype name	Col-0, Ler-0, Nipponbare, B73
`characteristics[growth condition]`	recommended	Growth conditions for the plant	pattern: Description of growth conditions	long day (16h light/8h dark), short day (8h light/16h dark), continuous light, greenhouse
`characteristics[treatment]`	recommended	*(override: requirement set to recommended)*		

14.9. clinical-metadata

Version: 1.0.0 | **Layer:** sample | **Extends:** sample-metadata | **Usable alone:** No

SDRF template for clinical study samples with treatment, demographics, and lifestyle metadata. Applicable to any organism. Combine with organism template (human, vertebrates) and technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[disease]`	required	*(override: requirement set to required)*		
`characteristics[compound]`	optional	Chemical compound or drug applied to sample	ontology: chebi, ncit, efo	doxorubicin, cisplatin, tamoxifen, metformin
`characteristics[dose]`	optional	Dose or concentration of compound treatment	number with unit (mg/kg, uM, nM, mg, ug, mg/mL, ug/mL, mM)	10 mg/kg, 50 uM, 100 nM, 5 mg
`characteristics[exposure duration]`	optional	Duration of treatment exposure	number with unit (hour, day, minute, week, month)	24 hour, 5 day, 30 minute, 2 week

Column Name	Req.	Description	Validators	Examples
`characteristics[treatment status]`	optional	Treatment status at time of sampling	values: pre-treatment, on treatment, post-treatment, treatment naive	
`characteristics[treatment response]`	optional	Response to treatment (for studies measuring therapeutic outcomes)	ontology: ncit	complete response, partial response, progressive disease, stable disease
`characteristics[pre-existing condition]`	optional	Pre-existing medical conditions or comorbidities	ontology: mondo, efo, doid	diabetes mellitus, hypertension, obesity
`characteristics[body mass index]`	optional	Body mass index (BMI) in kg/m^2	pattern: Numeric BMI value	24.5, 31.2, 18.7
`characteristics[smoking status]`	optional	Patient smoking status	ontology: ncit	never smoker, former smoker, current smoker
`characteristics[menopausal status]`	optional	Menopausal status for female patients	values: pre-menopausal, peri-menopausal, post-menopausal	
`characteristics[genetic modification]`	optional	Method of genetic modification (knockout, knockdown, overexpression, transduction)	ontology: efo	knockout, knockdown, overexpression, transduction, …
`characteristics[phenotype]`	optional	Observable characteristics or traits (drug sensitivity, molecular markers, expression phenotypes)	ontology: pato, efo	drug resistant, HER2-positive, high expresser, wild-type phenotype
`characteristics[weight]`	optional	Body weight of the subject	number with unit (kg, g, lb)	70 kg, 55 kg, 154 lb
`characteristics[height]`	optional	Height of the subject	number with unit (cm, m)	175 cm, 1.75 m, 160 cm
`characteristics[sampling site]`	optional	Specific anatomical location or context of sampling within the organism part	ontology: uberon, bto	tumor, normal tissue adjacent to tumor, left ventricle, frontal cortex

Column Name	Req.	Description	Validators	Examples
`characteristics[genotype]`	optional	Known genetic variant, mutation, or genotype of the subject	pattern: Genotype as free text (gene name + variant)	BRCA1 mutation carrier, KRAS G12D mutant, wild type, TP53 R175H

14.10. oncology-metadata

Version: 1.0.0 | **Layer:** sample | **Extends:** clinical-metadata | **Usable alone:** No

SDRF template for cancer/oncology study samples with tumor staging, grading, and clinical outcome metadata. Extends clinical-metadata with oncology-specific columns. Combine with organism template (human, vertebrates) and technology template (ms-proteomics, affinity-proteomics, or ms-metabolomics).

Column Name	Req.	Description	Validators	Examples
`characteristics[disease staging]`	optional	Disease progression stage (stage I-IV, chronic phase, end stage)	ontology: ncit, efo	stage I, stage II, stage III, stage IV, …
`characteristics[tumor grading]`	optional	Histological tumor grade (describes how abnormal cells look)	ontology: ncit	grade 1, grade 2, grade 3, grade 4, …
`characteristics[tumor stage]`	optional	TNM staging notation (describes extent of cancer spread)	ontology: ncit	T2N1M0, T3N0M0, T1N0M0, T4N2M1
`characteristics[tumor size]`	optional	Tumor size measurement	number with unit (cm, mm)	2.5 cm, 15 mm, 0.8 cm
`characteristics[tumor mass]`	optional	Tumor mass/weight measurement	number with unit (g, mg)	15 g, 250 mg
`characteristics[histologic subtype]`	optional	Cancer molecular or histologic subtype	ontology: ncit	luminal A, luminal B, HER2-enriched, triple-negative, …
`characteristics[metastasis site]`	optional	Location where cancer has spread from primary site	ontology: uberon, bto	liver, lung, bone, brain
`characteristics[biopsy site]`	optional	Specific anatomical location of biopsy	ontology: uberon, bto	breast, colon, prostate, lung
`characteristics[clinical data]`	optional	Free-text clinical details (receptor status, treatment history, surgical details)	pattern: Free-text clinical data	ER+/PR+/HER2-, prior chemotherapy with doxorubicin, surgical resection performed

Column Name	Req.	Description	Validators	Examples
`characteristics[clinical history]`	optional	Relevant medical history information for the patient	pattern: Free-text clinical history	family history of breast cancer, previous radiation therapy, no significant medical history
`characteristics[survival time]`	optional	Patient survival time for survival analysis studies	number with unit (month, year, day, week)	24 month, 3 year, 180 day
`characteristics[last follow up]`	optional	Time of last clinical follow-up for longitudinal studies	number with unit (month, year, day, week)	36 month, 5 year, 365 day
`characteristics[mitotic rate]`	optional	Number of mitoses per high-power field (indicator of tumor proliferation)	pattern: Mitotic rate as count or count per HPF	5, 12/10 HPF, 3/10 HPF
`characteristics[dukes stage]`	optional	Dukes staging for colorectal cancer (A, B, C, D)	values: A, B, C, D	
`characteristics[ann arbor stage]`	optional	Ann Arbor staging for lymphoma (I, II, III, IV with optional A/B suffix)	pattern: Ann Arbor stage (I-IV with optional A/B suffix for symptoms, E for extranodal, S for spleen)	IA, IIB, IIIA, IVB, …
`characteristics[gleason score]`	optional	Gleason score for prostate cancer grading (sum of two pattern grades, range 2-10)	pattern: Gleason score as sum (e.g., 7) or component pattern (e.g., 3+4)	7, 3+4, 4+3, 9, …
`characteristics[weiss grade]`	optional	Weiss scoring system for adrenal cortical carcinoma (low or high)	values: low, high	

14.11. dia-acquisition

Version: 1.1.0 | **Layer:** experiment | **Extends:** ms-proteomics | **Usable alone:** No

SDRF template for Data-independent acquisition (DIA) experiments. Extends ms-proteomics with DIA-specific columns.

Column Name	Req.	Description	Validators	Examples
`comment[proteomics data acquisition method]`	required	Mass spectrometry acquisition method (restricted to DIA for this template)	single value only; values: Data-independent acquisition	
`comment[scan window lower limit]`	recommended	Lower m/z limit of the DIA scan window	pattern: m/z value as a number	400, 350.5
`comment[scan window upper limit]`	recommended	Upper m/z limit of the DIA scan window	pattern: m/z value as a number	1200, 1000
`comment[isolation window width]`	recommended	Width of the isolation window in m/z units	pattern: Width in m/z	25, 8, 4
`comment[dia method]`	recommended	Specific DIA method variant used	ontology: pride	SWATH-MS, MSE, All ion fragmentation, diaPASEF

14.12. single-cell

Version: 1.0.0 | **Layer:** experiment | **Extends:** ms-proteomics | **Usable alone:** No

SDRF template for single-cell proteomics (SCP) experiments. Works with any organism - combine with appropriate sample template (human, vertebrates, invertebrates, or plants). Aligned with Nature Methods SCP guidelines (Gatto et al., 2023).

Column Name	Req.	Description	Validators	Examples
`characteristics[sample type]`	recommended	*(override: requirement set to recommended)*		
`characteristics[single cell isolation protocol]`	required	Method used to isolate single cells (FACS, cellenONE, LCM, etc.)	values: FACS, cellenONE, microfluidics, laser capture microdissection, …	
`characteristics[cell identifier]`	required	Unique identifier for each single cell within the experiment. Required per SCP guidelines for tracking cells through analysis.	identifier	cell_001, SC_A1, well_B3, barcode_ATCGATCG, …
`comment[sample preparation batch]`	recommended	Batch identifier for sample preparation (plate, chip, processing batch). Critical for batch effect correction.		

Column Name	Req.	Description	Validators	Examples
characteristics[cells per well]	recommended	Number of cells per well/reaction. Use 1 for true single cells, higher numbers for small pools.	pattern: Number of cells	1, 5, 10, 100
comment[carrier channel]	recommended	TMT/TMTpro channel used for the carrier proteome	pattern: TMT channel label for carrier	TMT131C, TMTpro134N, TMT126
comment[reference channel]	recommended	TMT/TMTpro channel used for the reference sample (for normalization across sets)	pattern: TMT channel label for reference	TMT131N, TMTpro133C, TMT127N
characteristics[forward scatter]	optional	Forward scatter (FSC) value from flow cytometry - proxy for cell size	pattern: FSC value (numeric)	316.0, 250
characteristics[side scatter]	optional	Side scatter (SSC) value from flow cytometry - proxy for cell granularity/complexity	pattern: SSC value (numeric)	301.0, 184
characteristics[enrichment marker]	optional	Markers used for cell sorting/enrichment with optional intensity values	pattern: Enrichment marker(s) and optional intensity	CD45+, GFP+, CD3+CD4+, CD34:APC-Cy7-A=276.0, …
characteristics[cell viability]	optional	Viability status of the cell at isolation	values: live, viable, dead, unknown	
characteristics[cell cycle phase]	optional	Cell cycle phase if determined (e.g., by FACS or computational inference)	values: G1, S, G2, G2/M, …	
characteristics[cell diameter]	optional	Physical diameter of the isolated cell if measured (in micrometers)	number with unit (um, ¬m)	15 um, 20.5 um, 12 ¬m
characteristics[spatial coordinates]	optional	X,Y coordinates if cells were isolated from a spatial context (e.g., LCM from tissue)	pattern: Spatial coordinates	X=100;Y=250, X=50.5;Y=120.3
comment[tissue section]	optional	Tissue section identifier for spatially resolved single-cell proteomics	pattern: Tissue section identifier	section_001, slide_A_section_3
comment[facs nozzle size]	optional	Nozzle diameter used for FACS-based single cell isolation (in micrometers)	number with unit (um, ¬m)	70 um, 100 um, 130 ¬m
comment[facs sorting mode]	optional	Sorting mode used during FACS isolation	values: single cell, purity, yield, 4-way purity	

Column Name	Req.	Description	Validators	Examples
`comment[microfluidics chip type]`	optional	Type and manufacturer of the microfluidics chip used for single cell isolation	pattern: Chip type/manufacturer identifier	Fluidigm C1, Cellenion cellenCHIP, nanowell chip
`comment[lcm microscope model]`	optional	Model of the laser capture microdissection microscope used for cell isolation	pattern: LCM microscope model name	Leica LMD7, Zeiss PALM MicroBeam, Thermo LCM
`comment[nanopots chip version]`	optional	Version of the nanoPOTS chip used for single cell sample preparation	pattern: nanoPOTS chip version identifier	nanoPOTS v1, nanoPOTS v2, 9-well chip

14.13. immunopeptidomics

Version: 1.0.0 | **Layer:** experiment | **Extends:** ms-proteomics | **Usable alone:** No

SDRF template for immunopeptidomics experiments (MHC-bound peptide identification). Works with any organism - combine with appropriate sample template (human for HLA typing, vertebrates for H-2/MHC typing in mouse, etc.).

Column Name	Req.	Description	Validators	Examples
`characteristics[mhc protein complex]`	required	MHC protein complex targeted for immunopeptidome enrichment (GO:0042611)	values: MHC class I protein complex, MHC class II protein complex, non-classical MHC protein complex, mutant MHC protein complex, MHC protein complex with serotype	
`characteristics[immunopeptidome enrichment method]`	required	Method used to enrich MHC-bound peptides	values: immunoaffinity purification, immunoaffinity purification (iodoacetamide), mild acid elution, detergent lysis	

Column Name	Req.	Description	Validators	Examples
`characteristics[mhc typing]`	recommended	MHC alleles expressed by the sample (PRIDE:0000893) following IPD-MHC nomenclature (https://www.ebi.ac.uk/ipd/mhc/). Use IPD-IMGT/HLA notation for human (HLA-A*02:01), H-2 notation for mouse (H-2Kb, H-2Db), or appropriate IPD-MHC notation for other species. Multiple alleles can be separated by semicolons.	pattern: MHC allele notation (HLA for human, H-2 for mouse). Supports multi-allele (semicolon-separated), 2-4 field resolution.	HLA-A*02:01, HLA-B*07:02, HLA-A*02:01;HLA-B*07:02;HLA-C*07:02, HLA-A*02:01:01, …
`characteristics[mhc typing method]`	optional	MHC typing method used (PRIDE:0000894). Values mapped to NCIT where available: NGS-based typing (NCIT:C101293), sequence-based typing (NCIT:C130180), PCR-SSO (NCIT:C130181), PCR-SSP (NCIT:C130179), PCR-based genotyping (NCIT:C17003)	values: NGS-based typing, sequence-based typing, PCR-SSO, PCR-SSP, …	
`characteristics[antibody enrichment]`	recommended	Antibody clone used for MHC immunoprecipitation	pattern: Antibody clone name	W6/32, BB7.2

14.14. crosslinking

Version: 1.0.0 | **Layer:** experiment | **Extends:** ms-proteomics | **Usable alone:** No

SDRF template for crosslinking mass spectrometry (XL-MS) experiments. Extends ms-proteomics with crosslinking-specific columns for data analysis.

Column Name	Req.	Description	Validators	Examples
`comment[chemical cross-linking coupled with ms]`	recommended	MS-based cross-linking methodology used to identify this as a crosslinking dataset	values: cross-linking mass spectrometry	
`characteristics[enrichment process]`	recommended	*(override: requirement set to recommended)*		

Column Name	Req.	Description	Validators	Examples
comment[cross-linker]	required	Cross-linker compound with structured properties for analysis tools. Format: NT=name;AC=accession;CL=cleavable;TA=targets;MH/ML=stub masses Uses XLMOD ontology (parent term XLMOD:00004).	structured_kv	NT=DSS;AC=XLMOD:02001, NT=BS3;AC=XLMOD:02000, NT=DSSO;AC=XLMOD:02010;CL=yes;TA=K,S,T,Y,nterm;MH=54.01;ML=85.98, NT=EDC;AC=XLMOD:02009;CL=no;TA=K,D,E
comment[dissociation method]	required	Fragmentation method used in MS2. Critical for cleavable crosslinkers (DSSO, DSBU) which generate diagnostic stubions under specific fragmentation conditions.	ontology: ms, pride	HCD, CID, ETD, EThcD, …
comment[collision energy]	recommended	Collision energy used for fragmentation. Important for cleavable crosslinker analysis.	pattern: Collision energy format: {value} {unit} where unit is NCE or eV. For stepped collision energies, use semicolon-separated values or 'stepped' prefix.	30 NCE, 30% NCE, 27 eV, 25 NCE;27 NCE;30 NCE, …
comment[crosslink enrichment method]	recommended	Method used to enrich crosslinked peptides before MS analysis	ontology: pride, ms	size exclusion chromatography, strong cation exchange chromatography, high-pH reversed-phase chromatography, FAIMS
characteristics[crosslink distance]	optional	Maximum C¬-C¬ distance constraint provided by the crosslinker (for structural interpretation)	number with unit (Å)	30 Å, 26.4 Å, 11.4 Å
comment[crosslinker concentration]	optional	Concentration of crosslinking reagent used	number with unit (mM, uM, µM)	2 mM, 500 uM, 1 mM

Column Name	Req.	Description	Validators	Examples
`characteristics[crosslinking reaction time]`	optional	Duration of the crosslinking reaction	number with unit (min, h, s)	30 min, 1 h, 45 min
`characteristics[crosslinking temperature]`	optional	Temperature at which crosslinking was performed	number with unit (°C)	25°C, 4°C, 37°C, room temperature
`comment[crosslinker to protein ratio]`	optional	Molar ratio of crosslinker to protein	pattern: Ratio format (e.g., 50:1 or 1:1 w/w)	3001, 6001, 1:1 w/w
`comment[quenching reagent]`	optional	Reagent used to quench the crosslinking reaction	pattern: Chemical name of quenching reagent	Tris-HCl, ammonium bicarbonate, glycine

14.15. cell-lines

Version: 1.1.0 | **Layer:** experiment | **Extends:** sample-metadata | **Usable alone:** No

SDRF template for cell line samples with Cellosaurus-based annotation. Cell lines can originate from any organism - combine with appropriate organism template (human for HeLa, vertebrates for NIH 3T3, invertebrates for Sf9).

Column Name	Req.	Description	Validators	Examples
`characteristics[cell line]`	required	Name of the cell line	ontology: clo, bto, efo	HeLa, HEK293, MCF7, A549
`characteristics[disease]`	required	Disease state of the donor tissue from which the cell line was established		
`characteristics[cellosaurus accession]`	required	Cellosaurus accession number for the cell line	accession: cellosaurus	CVCL_0030, CVCL_0004
`characteristics[cellosaurus name]`	recommended	Official Cellosaurus name for the cell line		
`characteristics[sampling site]`	optional	Original source term, local provenance, or finer tissue-of-origin context for the cell line	ontology: uberon, bto	cervix, kidney, breast
`characteristics[passage number]`	recommended	Passage number of the cell line used in the experiment	pattern: Passage number should be an integer or range	10, 15-20, 5
`characteristics[biorepository]`	optional	BioBank or source from which the cell line was obtained	pattern: Source of the cell line	ATCC, DSMZ, ECACC, Sigma-Aldrich

Column Name	Req.	Description	Validators	Examples
`characteristics[cell line authentication]`	optional	Method used to authenticate the cell line identity	pattern: Authentication method used	STR profiling, SNP fingerprinting, cytogenetic analysis
`characteristics[culture medium]`	recommended	Culture medium used to grow the cell line	ontology: ncit	DMEM, RPMI 1640, MEM, Ham's F-12
`characteristics[developmental stage]`	optional	Developmental stage of the donor from which the cell line was derived	ontology: efo	adult, embryonic, fetal, neonatal
`characteristics[ancestry category]`	optional	Ancestry category of the cell line donor (if known)	ontology: hancestro	European, African, East Asian, South Asian
`characteristics[sample storage temperature]`	recommended	Storage temperature of the cell line (in Celsius)	number with unit (°C)	-80 °C, -20 °C, 4 °C

14.16. metaproteomics

Version: 1.0.0 | **Layer:** sample | **Extends:** base | **Usable alone:** No

Base SDRF template for metaproteomics experiments (microbial community proteomics). Extends base directly and defines MIxS-aligned sample metadata. When combined with ms-proteomics, sample-metadata columns (organism, disease, cell type) are excluded. Use a child template (human-gut, soil, water) for environment-specific fields.

Column Name	Req.	Description	Validators	Examples
`characteristics[environmental sample type]`	required	Type of environmental sample analyzed (ENVO or EFO term). Corresponds to MIxS env_medium (MIXS:0000014).	ontology: envo, efo	soil, seawater, gut microbiome, wastewater, …
`characteristics[geographic location]`	recommended	Geographic location where sample was collected (GAZ term or coordinates). Corresponds to MIxS geo_loc_name (MIXS:0000010).	ontology: gaz	Pacific Ocean, Amazon rainforest, 47.6062 N, 122.3321 W
`characteristics[environmental medium]`	recommended	Environmental material from which the sample was obtained (ENVO term). Corresponds to MIxS env_medium (MIXS:0000014).	ontology: envo	soil, seawater, freshwater, feces, …
`characteristics[collection date]`	optional	Date when sample was collected (ISO 8601)	date	2024, 2024-01, 2024-01-15

Column Name	Req.	Description	Validators	Examples
`characteristics[sample collection method]`	optional	Method used to collect the environmental sample	pattern: Collection method description	grab sample, core sample, swab, filtration
`characteristics[depth]`	optional	Depth at which sample was collected. Corresponds to MIxS depth (MIXS:0000018).	number with unit (m, cm, mm)	10 m, 50 cm, 100 m
`characteristics[altitude]`	optional	Altitude or elevation of sampling site. Corresponds to MIxS elevation (MIXS:0000093).	number with unit (m)	500 m, 1200 m, 0 m
`characteristics[temperature]`	optional	Temperature at sampling location. Corresponds to MIxS temperature (MIXS:0000113).	number with unit (°C)	25 °C, 4 °C, -20 °C
`characteristics[ph]`	optional	pH at sampling location	pattern: pH value	7.0, 5.5, 8.2
`characteristics[sample storage]`	optional	Storage conditions for the sample before analysis	pattern: Storage conditions	-80C, liquid nitrogen, 4C
`comment[metagenome accession]`	optional	Accession number for matched metagenome data	accession:	MGYA00001234, SRP123456
`characteristics[microbiome source]`	optional	Source of the microbiome being studied (e.g., gut microbiome, rhizosphere microbiome)	pattern: Microbiome source description	gut microbiome, rhizosphere microbiome, oral microbiome, skin microbiome
`characteristics[biomass estimation]`	optional	Estimated microbial biomass in the sample	pattern: Biomass estimation	1e9 cells/g, high biomass, low biomass
`characteristics[host contamination]`	optional	Level of host protein contamination if known	pattern: Host contamination level	low (<5%), moderate (5-20%), high (>20%)
`comment[contaminant database]`	optional	Contaminant database(s) used in database search	pattern: Contaminant database name(s)	cRAP, MaxQuant contaminants, cRAP;MaxQuant contaminants
`characteristics[mock community]`	optional	Identifier or name of mock community standard used	pattern: Mock community identifier	ZymoBIOMICS Microbial Community Standard, ATCC MSA-1000

Column Name	Req.	Description	Validators	Examples
`characteristics[mock community composition]`	optional	Description of mock community composition (species and ratios)	pattern: Community composition description	8 bacteria + 2 yeasts at defined ratios, even mix of 10 species
`comment[expected organism list]`	optional	Semicolon-separated list of organisms expected in mock community	pattern: Semicolon-separated organism list	E. coli;B. subtilis;S. cerevisiae;L. fermentum, Bacillus subtilis;Staphylococcus aureus

14.17. human-gut

Version: 1.0.0 | **Layer:** sample | **Extends:** metaproteomics | **Usable alone:** No

SDRF template for human gut metaproteomics. Extends metaproteomics with host-associated columns aligned with the GSC MIxS human-gut extension (0016004). Combine with ms-proteomics for MS acquisition columns.

Column Name	Req.	Description	Validators	Examples
`source name[sample name]`	required	Reference to the Sample. This term is used for cross-reference samples in SDRF for proteomics (MIXS:0001107).	identifier	Sample_1, Sample_2
`characteristics[host organism]`	required	Host organism for host-associated microbiome samples	ontology: ncbitaxon	Homo sapiens
`characteristics[host subject id]`	recommended	De-identified unique identifier for the host subject. Corresponds to MIxS host_subject_id (MIXS:0000861).	identifier	subject_001, patient_A, anonymized
`characteristics[host disease status]`	recommended	Host disease diagnoses. Corresponds to MIxS host_disease_stat (MIXS:0000031).	ontology: mondo, doid	inflammatory bowel disease, colorectal cancer, healthy
`characteristics[host body site]`	recommended	Body site where sample was obtained. Corresponds to MIxS host_body_site (MIXS:0000867).	ontology: uberon, bto	stool, oral cavity, colon
`characteristics[host genotype]`	recommended	Observed genotype MIXS:0000365.	ontology: efo, gene	homozygous, heterozygous
`characteristics[host phenotype]`	recommended	Phenotype of human or other host MIXS:0000274.	ontology: pato, hp	polydactyly, arachnodactyly

Column Name	Req.	Description	Validators	Examples
`characteristics[host age]`	optional	Age of host at the time of sampling. Corresponds to MIxS host_age (MIXS:0000255).	pattern: Age in standard format (Y=year, M=month, W=week, D=day, H=hour)	45Y, 8W, 3M
`characteristics[host sex]`	optional	Sex of the host organism. Corresponds to MIxS host_sex (MIXS:0000811).	values: male, female, intersex	
`characteristics[host body-mass index]`	optional	Body mass index (weight/height^2). Corresponds to MIxS host_body_mass_index (MIXS:0000317).	pattern: BMI numeric value	22.5, 30.1, 18.5
`characteristics[host height]`	optional	Height of the host. Corresponds to MIxS host_height (MIXS:0000264).	number with unit (cm, m)	175 cm, 1.75 m
`characteristics[host total mass]`	optional	Total mass of the host. Corresponds to MIxS host_tot_mass (MIXS:0000263).	number with unit (kg, g)	70 kg, 85 kg
`characteristics[ethnicity]`	optional	Ethnicity of the host. Corresponds to MIxS ethnicity (MIXS:0000895).	pattern: Ethnicity description	European, East Asian, African
`characteristics[host diet]`	optional	Diet type of the host. Corresponds to MIxS host_diet (MIXS:0000869).	pattern: Diet description	omnivore, vegan, western diet, high-fiber
`characteristics[special diet]`	optional	Special dietary restrictions. Corresponds to MIxS special_diet (MIXS:0000905).	pattern: Special diet description	gluten-free, low FODMAP, ketogenic
`characteristics[host last meal]`	optional	Content of last meal and time since feeding. Corresponds to MIxS host_last_meal (MIXS:0000870).	pattern: Last meal description	breakfast 4 hours prior, fasting 12 hours
`characteristics[host family relationship]`	optional	Relationships to other hosts in the same study; can include multiple relationships MIXS:0000872.	pattern: family relationship description	father, daughter
`characteristics[host occupation]`	optional	Most frequent job performed by subject MIXS:0000896.	pattern: occupation description	teacher, engineer

Column Name	Req.	Description	Validators	Examples
characteristics[gastrointestinal tract disorder]	optional	History of GI tract disorders. Corresponds to MIxS gastroint_disord (MIXS:0000280).	pattern: GI disorder description	Crohn's disease, ulcerative colitis, irritable bowel syndrome, none
characteristics[liver disorder]	optional	History of liver disorders. Corresponds to MIxS liver_disord (MIXS:0000282).	pattern: Liver disorder description	none, fatty liver disease, hepatitis
characteristics[antibiotic treatment]	optional	Recent antibiotic exposure of the host	pattern: Antibiotic treatment description	none, amoxicillin 7 days prior, broad-spectrum
characteristics[ihmc medication code]	optional	Medication codes (IHMC). Corresponds to MIxS ihmc_medication_code (MIXS:0000884).	pattern: Medication code(s)	none, A02BC01, N02BE01
characteristics[host body product]	optional	Substance produced by the body where sample was obtained. Corresponds to MIxS host_body_product (MIXS:0000888).	pattern: Body product description	stool, mucus, saliva
characteristics[host body temperature]	optional	Core body temperature at sample collection. Corresponds to MIxS host_body_temp (MIXS:0000874).	number with unit (°C)	36.6 °C, 37.2 °C
characteristics[perturbation]	optional	Type of perturbation applied. Corresponds to MIxS perturbation (MIXS:0000754).	pattern: Perturbation description	antibiotic administration, dietary intervention, none
characteristics[chemical administration]	optional	Chemical compounds administered to the host. Corresponds to MIxS chem_administration (MIXS:0000751).	pattern: Chemical administration description	metformin 500mg daily, probiotics
characteristics[host pulse]	optional	Resting pulse, measured as beats per minutes (MIXS:0000333).	number with unit (bpm)	65 bpm, 72 bpm, 85 bpm
characteristics[observed host symbionts]	optional	The taxonomic name of the organism(s) found living in mutualistic, commensalistic, or parasitic symbiosis with the specific host (MIXS:0001298).	ontology:	

Column Name	Req.	Description	Validators	Examples
`characteristics[medical history performed]`	optional	Whether full medical history was collected (MIXS:0000897).	pattern: Medical history performed description	45 year old male with a 2 week history of stomach pain.
`characteristics[project name]`	optional	Name of the project within which the sequencing was organized (MIXS:0000092).	identifier	Human_microbiome_project
`comment[sample storage temperature]`	optional	Temperature at which sample was stored (MIXS:0000110).	number with unit (°C)	-80 °C, -20 °C
`comment[sample storage location]`	optional	Location at which sample was stored (MIXS:0000755).	pattern: storage location description	freezer, room
`comment[sample storage duration]`	optional	Duration for which the sample was stored (MIXS:0000116).	pattern: storage duration description	40 days, 6 months
`comment[sample volume or weight for DNA extraction]`	optional	Volume (ml) or mass (g) of total collected sample processed for DNA extraction (MIXS:0000111).	number with unit (ml,g)	1 ml, 2 g
`comment[organism count]`	optional	Total cell count of any organism per gram, volume, or area of sample, including count (MIXS:0000103).	number with unit (CFU/ml)	500 CFU/ml, 2000 CFU/ml
`comment[oxygenation status of sample]`	optional	Oxygenation status of sample (MIXS:0000753).	number with unit (mmHg)	80 mmHg, 100 mmHg
`comment[miscellaneous parameter]`	optional	Any other measurement performed or parameter collected, that is not listed here (MIXS:0000752).	pattern: miscellaneous parameter description	none

14.18. soil

Version: 1.0.0 | **Layer:** sample | **Extends:** metaproteomics | **Usable alone:** No

SDRF template for soil metaproteomics. Extends metaproteomics with soil-specific columns aligned with the GSC MIxS soil extension (0016012). Combine with ms-proteomics for MS acquisition columns.

Column Name	Req.	Description	Validators	Examples
source name[sample name]	required	Reference to the Sample. This term is used for cross-reference samples in SDRF for proteomics (MIXS:0001107).	identifier	Sample_1, Sample_2
project name	required	Name of the project within which the sequencing was organized (MIXS:0000092).	identifier	Bermuda_Atlantic_Time-series_Study, Metaproteomes_of_Seawater_Microbial_Communities
characteristics[soil type]	recommended	Soil classification type (ENVO term)	ontology: envo	sandy loam, clay, peat, silt
characteristics[elevation]	optional	Height of the sampling site above mean sea level.	values: 10 feet, 30 feet, 2 cm	
characteristics[slope gradient]	optional	The angle between ground surface and a horizontal line (MIxS:0000646).	values: 10 %, 30 %	
characteristics[slope aspect]	optional	The direction a slope faces (MIxS:0000647).	values: 300 degrees, north west, south	
characteristics[profile position]	optional	Cross-sectional position in the hillslope where sample was collected (MIxS:0001084).	values: summit, shoulder, backslope	
characteristics[drainage classification]	optional	Drainage classification from a standard system such as the USDA system (MIxS:0001085).	values: well drained, poorly drained, very poorly drained	
characteristics[current land use]	optional	Present state of sample site (MIXS:0001080).	pattern: Land use type	agricultural, forest, urban, grassland, …
characteristics[current vegetation]	optional	Current vegetation type at sampling site. Corresponds to MIxS cur_vegetation (MIXS:0000312).	ontology: envo	grassland, deciduous forest, cropland
comment[current vegetation method]	optional	method used in vegetation classification (MIXS:0000314).	values: vegetation survey, quadrant sampling	
characteristics[history of previous land use]	optional	Previous land use and dates (MIXS:0000315).	values: agricultural land 15 years, grassland forest 5 years	

Column Name	Req.	Description	Validators	Examples
`comment[history of previous land use method]`	optional	Reference or method used in determining previous land use (MIXS:0000316).	values: land registry records, historical archives	
`characteristics[crop rotation]`	optional	Crop rotation history. Corresponds to MIxS crop_rotation (MIXS:0000318).	pattern: Crop rotation description	corn-soybean rotation, wheat-fallow, continuous corn
`characteristics[history of agrochemical additions]`	optional	Addition of fertilizers, pesticides, etc. - amount and time of applications (MIXS:0000639).	values: urea 2mg/L, 2020-01-01, superphosphate 5mg/L, 2019-02-19	
`characteristics[history of tillage]`	optional	tilling methods (MIXS:0001081).	values: chisel, digging spade	
`characteristics[history of fire]`	optional	evidence of fire (MIXS:0001086).	date	09-10-2025, 10-02-2002
`characteristics[history of flooding]`	optional	evidence of flooding (MIXS:0000319).	date	09-10-2025, 10-02-2002
`characteristics[history of extreme event]`	optional	events that may have affected microbial populations (MIXS:0000320).	date	09-10-2025, 10-02-2002
`characteristics[soil horizon]`	optional	Soil horizon from which sample was collected	values: O horizon, A horizon, B horizon, C horizon, …	
`comment[horizon method]`	optional	method used in determining the horizon (MIXS:0000321).	pattern: Horizon method description	USDA Field Book, FAO Guidelines for Soil Profile Description, none
`characteristics[soil taxonomic of FAO classification]`	optional	Soil classification from the FAO World soil distribution (MIxS:0001083)	values: acrisols, calcisols, durisols, gleysols	
`characteristics[link classification information]`	optional	Link to digitized soil maps or other soil classification information (MIXS:0000329).	pattern: URL, DOI, or PMID linking to classification information	https://example.com/soil-map, doi:10.1000/xyz123, PMID:12345678
`characteristics[soil taxonomic local classification]`	optional	Soil classification based on local soil classification system (MIXS:0000330).	pattern: Local soil classification term or description	Chernozem, Podzol, Local classification type A

Column Name	Req.	Description	Validators	Examples
`characteristics[soil local classification method]`	optional	Method used for local soil classification, including reference to the classification system or methodology applied (MIXS:0000331).	pattern: Description or reference to the local soil classification method	FAO soil classification guidelines, USDA Soil Taxonomy methodology, National soil classification system (UK)
`characteristics[soil texture]`	optional	The relative proportion of different grain sizes of mineral particles in a soil, expressed as percentages of sand, silt, and clay, optionally including a textural class (MIXS:0000335).	pattern: Soil texture expressed as percentages and/or textural class	40% sand 40% silt 20% clay, 30-50 silty clay loam, silty clay loam
`characteristics[soil texture method]`	optional	Method used to determine soil texture, including reference to the analytical or classification approach applied.(MIXS:0000336)	pattern: Description or reference to the soil texture determination method	Hydrometer method, Laser diffraction particle size analysis, Pipette method
`characteristics[link climate information]`	optional	Link to climate resource associated with the sampling site. (MIXS:0000328)	pattern: URL, DOI, or PMID linking to climate information	https://climate-data.org/location123, doi:10.1000/climate456, PMID:98765432
`characteristics[mean annual temperature]`	optional	Mean annual temperature at the sampling site, typically expressed in degrees Celsius.(MIXS:0000135)	values:	12.5 oC, 25 oC
`characteristics[mean seasonal temperature]`	optional	Mean seasonal temperature range or average during a defined season at the sampling site. (MIXS:0000136)	pattern: Seasonal temperature description or value	summer 18°C, winter -2 to 5°C
`characteristics[mean annual precipitation]`	optional	Total annual precipitation at the sampling site, typically measured in millimeters per year.(MIXS:0000140)	numeric	800/year, 1200.5/year
`characteristics[mean seasonal precipitation]`	optional	Precipitation amount during a defined season at the sampling site. (MIXS:0000141)	pattern: Seasonal precipitation description or value	summer 120 mm, winter 300 mm

Column Name	Req.	Description	Validators	Examples
`characteristics[pH method]`	optional	Method used to determine pH in the sample.(MIXS:0001106)	pattern: Description of pH measurement method	pH measured in water (1:2.5 soil:solution ratio), pH measured in CaCl2 solution
`characteristics[organic matter]`	optional	Organic matter content of the sample.(MIXS:0000204)	number with unit ()	5.2 %, 12 %
`characteristics[total organic carbon]`	optional	Total organic carbon content. Corresponds to MIxS tot_org_carb (MIXS:0000533).	pattern: Total organic carbon with unit	15.2 g/kg, 2.5 %
`characteristics[total organic carbon method]`	optional	Method used to determine total organic carbon.(MIXS:0000337)	pattern: Description of TOC measurement method	dry combustion method, Walkley-Black method
`characteristics[organic nitrogen]`	optional	Organic nitrogen content of the sample.(MIXS:0000205)	number with unit ()	0.2 %, 1.5 %
`characteristics[total nitrogen content]`	optional	Total nitrogen content of the sample.(MIXS:0000530)	number with unit ()	0.5 %, 2.1 %
`characteristics[total nitrogen content method]`	optional	Method used to determine total nitrogen content.(MIXS:0000338)	pattern: Description of nitrogen measurement method	Kjeldahl method, Dumas combustion method
`characteristics[water content]`	optional	Water content of soil sample. Corresponds to MIxS water_content (MIXS:0000185).	pattern: Water content with unit	25 %, 0.25 g/g
`characteristics[soil water content method]`	optional	Method used to determine soil water content.(MIXS:0000323)	pattern: Description of water content measurement method	gravimetric method, time-domain reflectometry
`comment[microbial biomass]`	optional	Microbial biomass present in the sample.(MIXS:0000650)	number with unit ()	250 mg/kg, 1200 mg/kg
`comment[microbial biomass method]`	optional	Method used to determine microbial biomass.(MIXS:0000339)	pattern: Description of microbial biomass measurement method	chloroform fumigation extraction, substrate-induced respiration

Column Name	Req.	Description	Validators	Examples
`characteristics[extreme unusual properties of heavy metals]`	optional	Concentration of heavy metals in the sample, including elements such as lead, cadmium, mercury, and others.(MIXS:0000652)	pattern: Heavy metal concentrations with units or description	Pb 50 mg/kg; Cd 2 mg/kg, Hg 0.5 mg/kg
`characteristics[extreme unusual properties of heavy metals method]`	optional	Method used to determine heavy metal concentrations in the sample.(MIXS:0000343)	pattern: Description of heavy metal measurement method	ICP-MS analysis, Atomic absorption spectroscopy
`characteristics[aluminum saturation]`	optional	Aluminum saturation of the soil, typically expressed as a percentage of cation exchange capacity occupied by aluminum.(MIXS:0000607)	number with unit ()	15 %, 45.5 %
`characteristics[aluminum saturation method]`	optional	Method used to determine aluminum saturation in the sample.(MIXS:0000324)	pattern: Description of aluminum saturation measurement method	KCl extraction followed by titration, ICP-OES after extraction
`characteristics[sieving]`	optional	Size of mesh used to sieve the sample prior to analysis.(MIXS:0000322)	pattern: Sieve size or description	2 mm mesh, 500 µm sieve
`comment[sample volume or weight for DNA extraction]`	optional	Volume or weight of sample used for DNA extraction.(MIXS:0000111)	pattern: Sample amount with unit	0.5 g, 10 mL
`characteristics[pooling of DNA extracts]`	optional	Whether multiple DNA extracts were pooled prior to sequencing.(MIXS:0000325)	pattern: Indication of DNA extract pooling	yes, pooled from 3 replicates, no
`comment[storage conditions]`	optional	Storage conditions of the sample prior to analysis, including temperature and duration if known.(MIXS:0000327)	pattern: Storage condition description	-80°C for 6 months, 4°C short-term storage, room temperature
`comment[link to additional analysis]`	optional	Link to additional analysis results or external resources related to the sample.(MIXS:0000340)	pattern: URL, DOI, or PMID linking to additional analysis	https://example.com/analysis-results, doi:10.1000/analysis123, PMID:12345678
`comment[miscellaneous parameter]`	optional	Any additional parameter not captured by other fields; should include a name and value.(MIXS:0000752)	pattern: Parameter name and value in key:value format	{'enzyme_activity': 'high'}, {'soil_color': 'dark brown'}

14.19. water

Version: 1.0.0 | **Layer:** sample | **Extends:** metaproteomics | **Usable alone:** No

SDRF template for aquatic metaproteomics. Extends metaproteomics with water-specific columns aligned with the GSC MIxS water extension (0016014). Combine with ms-proteomics for MS acquisition columns.

Column Name	Req.	Description	Validators	Examples
`source name[sample name]`	required	Reference to the Sample. This term is used for cross-reference samples in SDRF for proteomics (MIXS:0001107).	identifier	Sample_1, Sample_2
`project name`	required	Name of the project within which the sequencing was organized (MIXS:0000092).	identifier	Bermuda_Atlantic_Time-series_Study, Metaproteomes_of_Seawater_Microbial_Communities
`characteristics[sampling depth zone]`	recommended	Ecological depth zone of the sampling site	values: epipelagic, mesopelagic, bathypelagic, abyssopelagic, …	
`characteristics[elevation]`	recommended	Height of the sampling site above mean sea level.	number with unit (feet, cm)	10 feet, 30 feet, 2 cm
`characteristics[tidal stage]`	recommended	Stage of tide (MIXS:0000750).	values: high tide, low tide	
`characteristics[total depth of water volume]`	recommended	Measurement of total depth of water column (MIXS:0000634).	number with unit (m)	35 m, 10 m, 200 m
`characteristics[water current]`	recommended	Measured speed and direction of water flow	number with unit (m/s)	0.35 m/s, 0.10 m/s
`characteristics[mean friction velocity]`	recommended	Measurement of mean friction velocity (MIXS:0000498).	number with unit (m/s)	0.035 m/s, 0.010 m/s
`characteristics[mean peak friction velocity]`	recommended	Measurement of mean peak friction velocity (MIXS:0000502).	number with unit (m/s)	0.015 m/s, 0.045 m/s
`characteristics[pressure]`	recommended	Pressure to which the sample is subject to, in atmospheres (MIXS:0000412)	number with unit (dbar)	15 dbar, 25 dbar
`characteristics[salinity]`	optional	Salinity measurement. Corresponds to MIxS salinity (MIXS:0000183).	pattern: Salinity value with unit or descriptive term	35 PSU, freshwater, brackish

Column Name	Req.	Description	Validators	Examples
characteristics[conductivity]	optional	Electrical conductivity of water sample. Corresponds to MIxS conduc (MIXS:0000544).	pattern: Conductivity with unit	450 uS/cm, 1.2 mS/cm
characteristics[density]	optional	Density of the sample, which is its mass per unit volume (aka volumetric mass density) (MIXS:0000435).	number with unit (kg/m3)	1022 kg/m3, 1050 kg/m3
characteristics[redox potential]	optional	Redox potential, measured relative to a hydrogen cell, indicating oxidation or reduction potential (MIXS:0000182).	number with unit (V)	1.2 V, 0.80 V
characteristics[fluorescence]	optional	Raw or converted fluorescence of water (MIXS:0000704).	number with unit (RFU)	1.2 RFU, 3.8 RFU
characteristics[light intensity]	optional	Light intensity at sampling depth. Corresponds to MIxS light_intensity (MIXS:0000706).	pattern: Light intensity with unit	500 lux, 100 umol/m2/s
characteristics[turbidity]	optional	Turbidity measurement. Corresponds to MIxS turbidity (MIXS:0000191).	pattern: Turbidity with unit	5.2 NTU, 12 FNU
characteristics[downward PAR]	optional	Visible waveband radiance and irradiance measurements in the water column (MIXS:0000703).	number with unit (µmol)	120 µmol, 15 µmol
characteristics[photon flux]	optional	Measurement of photon flux (MIXS:0000725).	number with unit (µmol)	120 µmol, 15 µmol
characteristics[nitrate]	optional	Nitrate concentration. Corresponds to MIxS nitrate (MIXS:0000425).	number with unit (mg/L, umol/L)	0.5 mg/L, 10 umol/L
characteristics[nitrite]	optional	Nitrite concentration. Corresponds to MIxS nitrite (MIXS:0000426).	number with unit (mg/L)	0.5 mg/L, 1 mg/L
characteristics[nitrogen]	optional	Nitrogen concentration. Corresponds to MIxS nitrogen (MIXS:0000504).	number with unit (mg/L)	10 mg/L, 15 mg/L
characteristics[ammonium]	optional	Ammonium concentration. Corresponds to MIxS ammonium (MIXS:0000427).	number with unit (mg/L)	1 mg/L, 1.5 mg/L
characteristics[phosphate]	optional	Phosphate concentration. Corresponds to MIxS phosphate (MIXS:0000505).	number with unit (mg/L, umol/L)	0.1 mg/L, 1.5 umol/L

Column Name	Req.	Description	Validators	Examples
characteristics[dissolved inorganic nitrogen]	optional	Dissolved inorganic nitrogen concentration (MIXS:0000698).	number with unit (mmol/L)	761 mmol/L, 500 mmol/L
characteristics[total inorganic nitrogen]	optional	total inorganic nitrogen concentration (MIXS:0000745).	number with unit (mg/L)	76 mg/L, 40 mg/L
characteristics[total dissolved nitrogen]	optional	total dissolved nitrogen concentration (MIXS:0000744).	number with unit (mg/L)	76 mg/L, 40 mg/L
characteristics[dissolved inorganic phosphorus]	optional	Dissolved inorganic phosphorus concentration (MIXS:0000106).	number with unit (mmol/L)	76 mmol/L, 56 mmol/L
characteristics[total nitrogen concentration]	optional	total nitrogen concentration (MIXS:0000102).	number with unit (mmol/L)	76 mmol/L, 56 mmol/L
characteristics[total phosphorus]	optional	total phosphorus concentration. Corresponds to MIxS total phosphorus (MIXS:0000117).	number with unit (mg/L)	0.03 mg/L, 0.05mg/L
characteristics[soluble reactive phosphorus]	optional	soluble reactive phosphorus concentration (MIXS:0000738).	number with unit (mg/L)	0.1 mg/L, 0.2 mg/L
characteristics[silicate]	optional	ssilicate concentration. Corresponds to MIxS(MIXS:0000184).	number with unit (mmol/L)	0.05 mmol/L, 0.08 mmol/L
characteristics[alkalinity]	optional	Alkalinity measurement. Corresponds to MIxS alkalinity (MIXS:0000421).	number with unit (mg/L, meq/L)	120 mg/L, 2.5 meq/L
characteristics[alkalinity method]	optional	Method used for alkalinity measurement (MIXS:0000298).	identifier	titration
characteristics[calcium]	optional	Calcium concentration. Corresponds to MIxS(MIXS:0000421).	number with unit (mmol/L)	0.2 mmol/L, 2.5 mmol/L
characteristics[magnesium]	optional	Magnesium concentration. Corresponds to MIxS(MIXS:0000431).	number with unit (mmol/kg)	25 mmol/kg, 52.8 mmol/kg
characteristics[potassium]	optional	Potassium concentration. Corresponds to MIxS(MIXS:0000430).	number with unit (mg/L)	463 mg/L, 300 mg/L

Column Name	Req.	Description	Validators	Examples
characteristics[sodium]	optional	Sodium concentration. Corresponds to MIxS(MIXS:0000428).	number with unit (mg/L)	10.5 mg/L, 12 mg/L
characteristics[chloride]	optional	Sodium concentration. Corresponds to MIxS(MIXS:0000429).	number with unit (mg/L)	5000 mg/L, 3500 mg/L
characteristics[sulfate]	optional	Sulfate concentration. Corresponds to MIxS(MIXS:0000423).	number with unit (mmol/L)	5 mmol/L, 3.5 mmol/L
characteristics[sulfide]	optional	Sulfide concentration. Corresponds to MIxS(MIXS:0000424).	number with unit (mmol/L)	2 mmol/L, 1.5 mmol/L
characteristics[bromide]	optional	Bromide concentration. Corresponds to MIxS(MIXS:0000176).	number with unit (ppm)	0.05 ppm, 0.01 ppm
characteristics[atmospheric data]	optional	Measurement of atmospheric data; can include multiple data (MIXS:0001097).	pattern: Diet description	wind speed; 9 knots, humidity; 75%
characteristics[dissolved organic carbon]	optional	Dissolved organic carbon concentration. Corresponds to MIxS (MIXS:0000433)	number with unit (mmol/L)	197 mmol/L, 75 mmol/L
characteristics[dissolved inorganic carbon]	optional	Dissolved inorganic carbon concentration. Corresponds to MIxS (MIXS:0000434)	number with unit (mmol/kg)	2059 mmol/kg, 1075 mmol/kg
characteristics[dissolved carbon dioxide]	optional	Dissolved carbon dioxide concentration. Corresponds to MIxS (MIXS:0000436)	number with unit (mg/L)	5 mg/L, 10 mg/L
characteristics[dissolved oxygen]	optional	Dissolved oxygen concentration. Corresponds to MIxS diss_oxygen (MIXS:0000119).	pattern: Dissolved oxygen with unit or descriptive term	8.5 mg/L, hypoxic, anoxic
characteristics[dissolved hydrogen]	optional	Dissolved hydrogen concentration. Corresponds to MIxS (MIXS:0000205).	number with unit (mmol/L)	0.3 mmol/L, 0.5 mmol/L
characteristics[dissolved organic nitrogen]	optional	Dissolved organic nitrogen concentration. Corresponds to MIxS (MIXS:0000162).	number with unit (mmol/L)	0.04 mmol/L, 0.05 mmol/L

Column Name	Req.	Description	Validators	Examples
`characteristics[organic carbon]`	optional	Organic carbon concentration. Corresponds to MIxS (MIXS:0000508).	number with unit (ug/L)	1.5 ug/L, 1 ug/L
`characteristics[organic matter]`	optional	Organic matter concentration. Corresponds to MIxS (MIXS:0000204).	number with unit (ug/L)	1.75 ug/L, 1 ug/L
`characteristics[organic nitrogen]`	optional	Organic nitrogen concentration. Corresponds to MIxS (MIXS:0000205).	number with unit (umol/L)	4 umol/L, 1 umol/L
`characteristics[particulate organic carbon]`	optional	Particulate organic carbon concentration. Corresponds to MIxS (MIXS:0000515).	number with unit (umol/L)	1.92 umol/L, 1 umol/L
`characteristics[particulate organic nitrogen]`	optional	Particulate organic nitrogen concentration. Corresponds to MIxS (MIXS:0000719).	number with unit (umol/L)	0.3 umol/L, 0.5 umol/L
`characteristics[total particulate carbon]`	optional	Total particulate carbon content. Corresponds to MIxS (MIXS:0000747).	number with unit (umol/L, ug/L)	35 umol/L, 5 ug/L
`characteristics[carbon nitrogen ratio]`	optional	Carbon to nitrogen concentration. Corresponds to MIxS (MIXS:0000310).	pattern: carbon nitrogen measurement value	0.475, 0.34576
`characteristics[suspended particulate matter]`	optional	Suspended particulate matter concentration. Corresponds to MIxS (MIXS:0000741).	number with unit (mg/L)	0.5 mg/L, 1 mg/L
`characteristics[chlorophyll]`	optional	Chlorophyll concentration if measured	number with unit (ug/L, mg/L)	2.5 ug/L, 0.1 mg/L
`characteristics[primary production]`	optional	Primary production measurement	number with unit (mg/m3/day, g/m2/day)	100 mg/m3/day, 10 g/m2/day
`comment[bacterial production]`	optional	bacterial production measurement (MIXS:0000683)	number with unit (mg/m3/day)	5 mg/m3/day, 10 mg/m3/day
`characteristics[bacterial respiration]`	optional	bacterial respiration measurement (MIXS:0000684)	number with unit (ug/m3/day, umolO2/L/hr)	300 umolO2/L/hr, 100 umolO2/L/hr
`characteristics[bacterial carbon production]`	optional	bacterial carbon production measurement (MIXS:0000173)	number with unit (ug/L/hr)	2.53 ug/L/hr, 1.23 ug/L/hr

Column Name	Req.	Description	Validators	Examples
characteristics[aminopeptidase activity]	optional	Aminopeptidase activity measurement (MIXS:0000172)	number with unit (mol/L/hr)	0.253 mol/L/hr, 0.123 mol/L/hr
characteristics[glucosidase activity]	optional	glucosidase activity measurement (MIXS:0000137)	number with unit (mol/L/hr)	5 mol/L/hr, 3 mol/L/hr
characteristics[phaeopigments]	optional	phaeopigments concentration (MIXS:0000180)	number with unit (mg/m3)	2.5 mg/m3, 3 mg/m3
characteristics[biomass]	optional	Total cell count of any organism (MIXS:0000174)	number with unit (ton, kg, g)	25 g, 3 kg
comment[organism count]	optional	phaeopigments concentration (MIXS:0000103)	number with unit (CFU/ml)	500 CFU/ml, 2000 CFU/ml
comment[oxygenation status of sample]	optional	Oxygenation status of sample (MIXS:0000753).	number with unit (mmHg)	80 mmHg, 100 mmHg
characteristics[alkyl diethers]	optional	Alkyl diethers concentration (MIXS:0000490).	number with unit (mol/L)	0.005 mol/L, 0.003 mol/L
characteristics[diether lipids]	optional	Diether lipids concentration (MIXS:0000178).	number with unit (ng/L)	0.5 ng/L, 0.3 ng/L
characteristics[bishomohopanol]	optional	Bishomohopanol concentration (MIXS:0000175).	number with unit (ug/L, ug/g)	14 ug/L, 4 ug/g
characteristics[phospholipid fatty acid]	optional	Phospholipid fatty acid concentration (MIXS:0000181).	pattern: Phospholipid fatty acid name and measurement value	Linoleic acid; 2.8 mg/L, Palmitic acid; 1.5 mg/L
characteristics[petroleum hydrocarbon]	optional	Petroleum hydrocarbon concentration (MIXS:0000516).	number with unit (umol/L)	0.05 umol/L, 0.02 umol/L
comment[sample storage temperature]	optional	Temperature at which sample was stored (MIXS:0000110).	number with unit (°C)	-80 °C, -20 °C
comment[sample storage location]	optional	Location at which sample was stored (MIXS:0000755).	pattern: storage location description	freezer, room
comment[sample storage duration]	optional	Duration for which the sample was stored (MIXS:0000116).	pattern: storage duration description	40 days, 6 months
comment[sample volume or weight for DNA extraction]	optional	Volume or weight of sample used for DNA extraction.(MIXS:0000111)	pattern: Sample amount with unit	0.5 g, 10 mL

Column Name	Req.	Description	Validators	Examples
`comment[size-fraction lower threshold]`	optional	mesh/pore size used to pre-filter/pre-sort the sample (MIXS:0000735).	number with unit (um)	0.1 um, 0.2 um
`comment[size-fraction upper threshold]`	optional	Mesh or pore size of the device used to retain the sample (MIXS:0000736).	number with unit (um)	10 um, 20 um
`characteristics[perturbation]`	optional	Type of perturbation applied. Corresponds to MIxS perturbation (MIXS:0000754).	pattern: Perturbation description	antibiotic administration, dietary intervention, none
`comment[miscellaneous parameter]`	optional	Any additional parameter not captured by other fields; should include a name and value.(MIXS:0000752)	pattern: Parameter name and value in key:value format	{'enzyme_activity': 'high'}, {'soil_color': 'dark brown'}

14.20. gc-ms-metabolomics

Version: 1.0.0-dev | **Layer:** experiment | **Extends:** ms-metabolomics | **Usable alone:** No

SDRF template for GC-MS-based metabolomics experiments. Extends ms-metabolomics with derivatization and GC-specific columns.

Column Name	Req.	Description	Validators	Examples
`technology type`	required	*(override: requirement set to required)*	single value only; values: GC-MS-based metabolomics	
`comment[ion source]`	required	*(override: requirement set to required)*	values: electron ionization, chemical ionization, electron impact ionization	
`comment[derivatization]`	required	Derivatization protocol applied to the sample prior to GC injection. Use `none` if no derivatization was performed (e.g. volatiles analysis).	pattern: Free-text derivatization protocol	MSTFA, MOX + MSTFA, BSTFA, methoximation followed by trimethylsilylation, …

Column Name	Req.	Description	Validators	Examples
`comment[derivatization agent]`	recommended	Chemical reagent(s) used for derivatization	ontology: chebi	N-methyl-N-(trimethylsilyl)trifluoroacetamide, N,O-bis(trimethylsilyl)trifluoroacetamide, methoxyamine hydrochloride
`comment[gc column]`	recommended	GC column used for separation, including stationary phase and dimensions	pattern: Free-text GC column description	DB-5MS 30 m x 0.25 mm x 0.25 um, HP-5MS 30 m x 0.25 mm x 0.25 um, Rxi-5Sil MS 30 m x 0.25 mm x 0.25 um
`comment[carrier gas]`	optional	Carrier gas used in the GC system	values: helium, hydrogen, nitrogen	
`comment[oven program]`	optional	Free-text description of the GC oven temperature program	pattern: Free-text oven program description	60C 1 min, ramp 10C/min to 300C, hold 5 min, 70C 2 min, ramp 5C/min to 320C, hold 10 min

14.21. lc-ms-metabolomics

Version: 1.0.0-dev | **Layer:** experiment | **Extends:** ms-metabolomics | **Usable alone:** No

SDRF template for LC-MS-based metabolomics experiments. Extends ms-metabolomics with chromatography-specific columns.

Column Name	Req.	Description	Validators	Examples
`technology type`	required	*(override: requirement set to required)*	single value only; values: LC-MS-based metabolomics	

Column Name	Req.	Description	Validators	Examples
`comment[chromatography type]`	required	Chromatography mode used for separation. Should be a CHMO term (e.g. CHMO:0002302 reversed-phase chromatography, CHMO:0001025 hydrophilic interaction chromatography).	pattern: Chromatography type as a CHMO-style ontology term (NT=…;AC=CHMO:NNNNNNN) or one of the common short names. Strict CHMO ontology validation is a follow-up.	NT=reversed-phase chromatography;AC=CHMO:0002302, NT=hydrophilic interaction chromatography;AC=CHMO:0001025, reversed-phase chromatography, hydrophilic interaction chromatography
`comment[chromatography column]`	recommended	LC column used for separation, including manufacturer, model, and dimensions	pattern: Free-text chromatography column description	Waters ACQUITY HSS T3 100 x 2.1 mm 1.8 um, Phenomenex Kinetex C18 150 x 2.1 mm 2.6 um, SeQuant ZIC-pHILIC 150 x 2.1 mm 5 um
`comment[mobile phase a]`	optional	Composition of mobile phase A	pattern: Free-text composition of mobile phase A	0.1% formic acid in water, 10 mM ammonium acetate in water pH 9.0, 5 mM ammonium formate in water
`comment[mobile phase b]`	optional	Composition of mobile phase B	pattern: Free-text composition of mobile phase B	0.1% formic acid in acetonitrile, 10 mM ammonium acetate in 90% acetonitrile pH 9.0, methanol
`comment[gradient]`	optional	Free-text description of the chromatographic gradient program	pattern: Free-text gradient description	5% B 0-1 min, ramp to 95% B at 15 min, hold 3 min, re-equilibrate, isocratic 50% B for 10 min, 0-100% B in 30 min
`comment[flow rate]`	optional	Mobile-phase flow rate	number with unit (nL/min, uL/min, mL/min)	300 nL/min, 250 uL/min, 0.4 mL/min

14.22. ms-metabolomics

Version: 1.0.0-dev | **Layer:** technology | **Extends:** sample-metadata | **Usable alone:** Yes

Base SDRF template for mass spectrometry-based metabolomics. This is the parent template for LC-MS and GC-MS metabolomics experiments.

Column Name	Req.	Description	Validators	Examples
`technology type`	required	Type of technology used	single value only; values: metabolite profiling by mass spectrometry, LC-MS-based metabolomics, GC-MS-based metabolomics	
`comment[instrument]`	required	Mass spectrometer instrument used	ontology: psi-ms, pride	Q Exactive, Xevo G2-S QTof, Orbitrap Fusion Lumos, timsTOF Pro
`comment[ion source]`	required	Ion source used for ionization (e.g. ESI, APCI, EI, MALDI). Should be a child of MS:1000008.	ontology: psi-ms	electrospray ionization, atmospheric pressure chemical ionization, electron ionization, matrix-assisted laser desorption ionization
`comment[scan polarity]`	required	Scan polarity (positive scan, negative scan, or polarity switching). Polarity switching is reserved for true in-method switching producing a single file; otherwise use two rows per sample (one per polarity).	single value only; values: positive scan, negative scan, polarity switching	
`comment[acquisition method]`	required	Mass spectrometry acquisition method (DDA, DIA, SIM, MRM, full scan)	ontology: pride	data-dependent acquisition, data-independent acquisition, selected reaction monitoring, parallel reaction monitoring, …

Column Name	Req.	Description	Validators	Examples
comment[metabolite assignment file]	recommended	Pointer to the metabolite assignment file (MAF) for this assay. Per-assay constant — the same value is repeated across all rows of the same assay. Mirrors the MetaboBank semantics of Comment[Metabolite Assignment File].	pattern: Path or filename of the MAF file	m_MTBLS1129_LC-MS_positive_reverse-phase_metabolite_profiling_v2_maf.tsv, maf_pos_lipidomics.tsv
comment[extraction method]	recommended	Sample extraction method used to obtain the metabolite fraction	pattern: Free-text extraction method, optionally CHMO-style NT=…;AC=CHMO:NNNNNNN	methanol-water extraction, MTBE extraction, Bligh-Dyer extraction, Folch extraction, …
comment[extraction solvent]	optional	Solvent or solvent mixture used for metabolite extraction	ontology: chebi	methanol, chloroform, methyl tert-butyl ether, acetonitrile
characteristics[analyte class]	recommended	Class of metabolites targeted by the extraction and acquisition method	values: polar metabolites, lipids, amino acids, fatty acids, …	
comment[internal standard]	optional	Internal standard(s) spiked into the sample for normalization or retention-time calibration	ontology: chebi	caffeine-d9, L-tryptophan-d5, palmitic acid-d31
characteristics[sample matrix]	recommended	Type of biological matrix used as input (e.g. serum, plasma, CSF, urine, tissue extract)	ontology: uberon, bto	serum, plasma, cerebrospinal fluid, urine, …
comment[sample preparation batch]	optional	Batch identifier for sample preparation (plate, chip, processing batch). Useful for batch-effect correction.	pattern: Sample preparation batch identifier	plate1, batch_20220601, prep_A
comment[lc batch]	optional	Liquid chromatography batch identifier for batch-effect tracking (e.g. column changes, LC system swaps)	pattern: LC batch identifier	LC1, column_A
comment[acquisition date]	optional	Date of MS data acquisition (ISO 8601 format recommended). Useful for tracking instrument drift and batch effects.	pattern: Acquisition date/time	2022-06-01, 2022-06-01T18:28:37

Column Name	Req.	Description	Validators	Examples
comment[ms2 mass analyzer]	optional	Mass analyzer used for MS2 acquisition	ontology: psi-ms	orbitrap, ion trap, TOF
comment[ms min mz]	optional	MS method-defined minimum precursor (MS1) m/z setting used to acquire the data	m/z value	100m/z, 200m/z
comment[ms max mz]	optional	MS method-defined maximum precursor (MS1) m/z setting used to acquire the data	m/z value	1200m/z, 1600m/z
comment[ms1 scan range]	optional	m/z scan range for MS1 spectra as an interval. Alternative to separate ms min mz / ms max mz columns.	m/z range interval	70m/z-1000m/z, 100m/z-1500m/z
comment[ms2 scan range]	optional	m/z scan range for MS2 spectra as an interval. Alternative to separate ms2 min mz / ms2 max mz columns.	m/z range interval	50m/z-1000m/z, 100m/z-1500m/z
comment[precursor mass tolerance]	optional	Precursor mass tolerance for feature detection / database search	number with unit (ppm, Da, mmu)	5 ppm, 10 ppm, 0.01 Da
comment[fragment mass tolerance]	optional	Fragment mass tolerance for feature detection / database search	number with unit (ppm, Da, mmu)	0.02 Da, 20 ppm
comment[collision energy]	optional	Collision energy used for fragmentation	pattern: Collision energy format: {value} {unit} where unit is NCE or eV. For multiple values, use semicolon-separated entries.	30 NCE, 25 eV, 25 NCE;27 NCE;30 NCE
comment[raw data file md5]	optional	MD5 integrity hash for the raw data file referenced in comment[data file]. Mirrors MetaboBank Comment[Raw Data File md5].	pattern: 32-character hex MD5 hash	d41d8cd98f00b204e9800998ecf8427e, 9e107d9d372bb6826bd81d3542a419d6
comment[processed data file]	optional	Pointer to the processed data file (e.g. peak table, feature matrix). Mirrors MetaboBank Comment[Processed Data File].	pattern: Path or filename of the processed data file	features_pos.tsv, peak_table_lipidomics.csv

Column Name	Req.	Description	Validators	Examples
`comment[processed data file md5]`	optional	MD5 integrity hash for the processed data file. Mirrors MetaboBank Comment[Processed Data File md5].	pattern: 32-character hex MD5 hash	d41d8cd98f00b204e9800998ecf8427e
`comment[metabolite assignment file md5]`	optional	MD5 integrity hash for the metabolite assignment file. Mirrors MetaboBank Comment[Metabolite Assignment File md5].	pattern: 32-character hex MD5 hash	d41d8cd98f00b204e9800998ecf8427e

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